GPS Research Slides

2026-xx-xx

Link

Template

- template

2026-04-07

Link

Our assumption

- We assume there is a “true rank for each solution~~everyone~~”, but there are some errors（for example, our judge’s perception ~~that person’s current condition or his preference~~）caused by human when making judgements.
  So that
  Final rank = True rank + error

Link

Gumbel distribution

- Type I extreme value distribution used to **model the maximum** (or minimum) of a number of samples.
- For example the maximum amount of rain within past 100 years.
- Because respondents are picking extremes, the unobserved random error in their evaluation is naturally modeled using the Gumbel distribution.
- So what is the proof that we can use this distribution?
  Additionally, both discrete choice experiment and best-worst scaling interpret the aforementioned choice process with the error terms following Gumbel distribution and obtain a Multinomial (or conditional) logit model.

Link

Temperature and noise

$$\tilde{q}_i = q_i + T \cdot \mathrm{Gumbel}(0, 1)$$

- What is temperature and noise?
  Temperature controls the degree of randomness in token selection. Lower temperatures are good for prompts that require a less open-ended or creative response, while higher temperatures can lead to more diverse or creative results. A temperature of 0 means that the highest probability tokens are always selected. In this case, responses for a given prompt are mostly deterministic, but a small amount of variation is still possible.

Link

Temperature and noise

$$\tilde{q}_i = q_i + T \cdot \text{Gumbel}(0, 1)$$

- What is temperature and noise?
  Gumbel Distributed Random Noise is a stochastic perturbation rooted in **Extreme Value Theory**, specifically designed to model the distribution of **maximum values**. In machine learning and data analysis, it is utilized to transform deterministic selection into probabilistic sampling, allowing models to explore discrete choices with controlled variability while maintaining statistical rigor through its unique asymmetrical, **heavy-tailed distribution**.

Link

Temperature and noise

$$\tilde{q}_i = q_i + T \cdot \text{Gumbel}(0, 1)$$

- The Gumbel-Max Trick with Temperature
- transform static numerical scores into stochastic (random) decisions

- q_i **(Logits / Base Score):** Represents the "intrinsic strength" or preference of an option (e.g., the raw weight of an airport improvement suggestion).
- $\text{Gumbel}(0, 1)$ **(Random Perturbation):** Noise drawn from a standard Gumbel distribution. it represents unobserved factors, latent individual biases, or random errors in real-world decision-making.
- T **(Temperature / Scaling Factor):** This is your "control knob" for randomness.
- $\tilde{q}_i$ **(Perturbed Score):** The final score after adding noise and adjusting for temperature.

Link

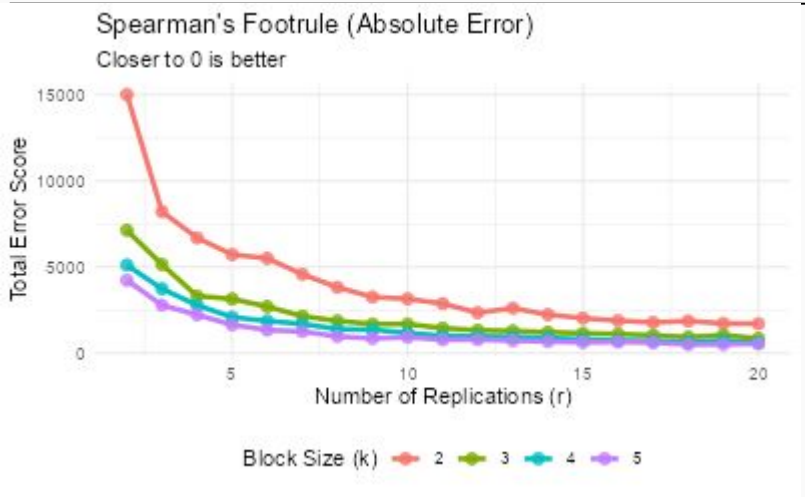
Temperature and noise

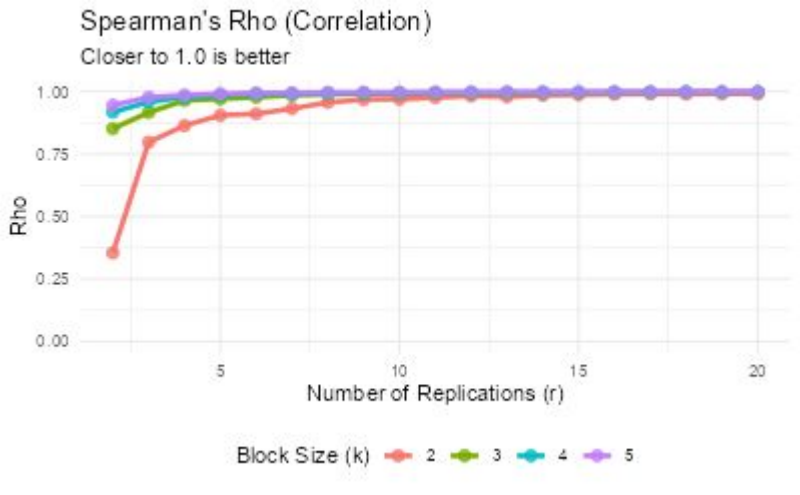
$$\tilde{q}_i = q_i + T \cdot \text{Gumbel}(0, 1)$$

- **Low T ($T \to 0$):** The noise is suppressed. The model becomes conservative and almost always picks the highest base score (Deterministic).
- **High T ($T \to \infty$):** The noise is amplified. The original scores (q_i) become irrelevant, and the choice becomes almost entirely random.

Link

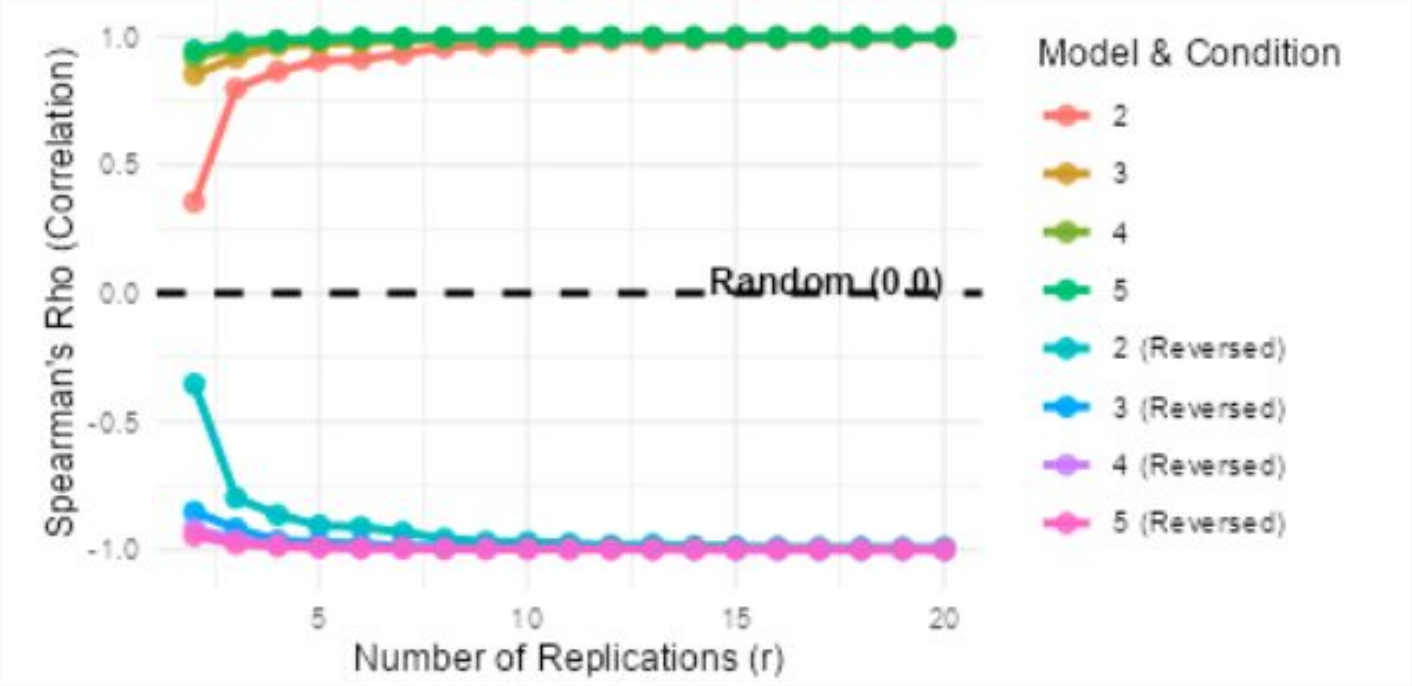
Footrule vs ρ





Link

Reverse ranking



Link

Q

- Should I mention the relationship of Gumbel & Random Utility Theory & logit model? ——yes
- 400 Airport Items -> Design Experiment: Alpha Design (cognitive load reduction) -> Collect Data: Best-Worst Scaling (BWS) -> Model Utility: Random Utility Theory (RUT) & Gumbel Assumption -> Estimate Ranking: Bradley-Terry (BT) Model

2026-03-31

Spearman's ρ

- What does it do？
  Calculate the correlation between 2 sets of ranking
  $\rho = 1 - \{ 6\Sigma d^2 / (n^3 - n)\}$, $-1 < \rho < 1$
- Where d = the differences between of the ranks of a treatment

treatment	Rank in 1st replication	Rank in 2nd replication	d
a	1	1	0
b	2	3	-1
c	3	2	1

Spearman’s footrule

- Calculate the absolute difference between 2 set of rankings
  $R = 1 - \{ 3\Sigma|d| / (n^2 - 1)\}$

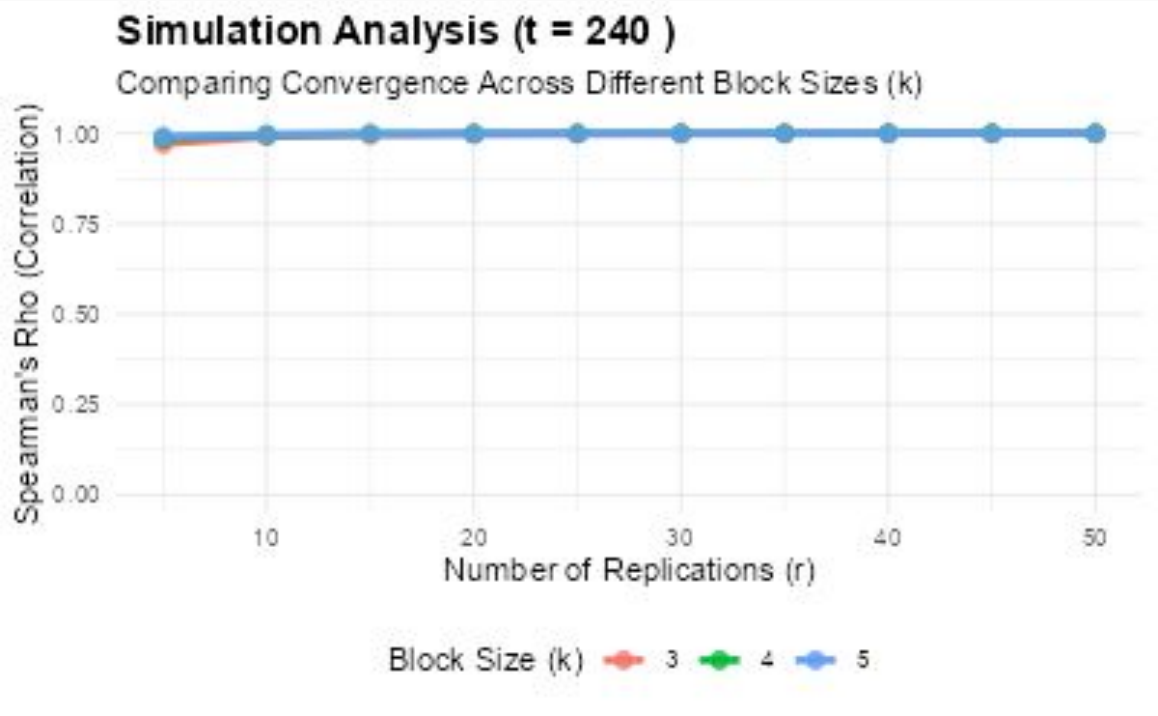
What can we use it to do

- Compare between 2 user segment

- Sample Stability Test (Split-Half Reliability) -> Check internal consensus within your sample group

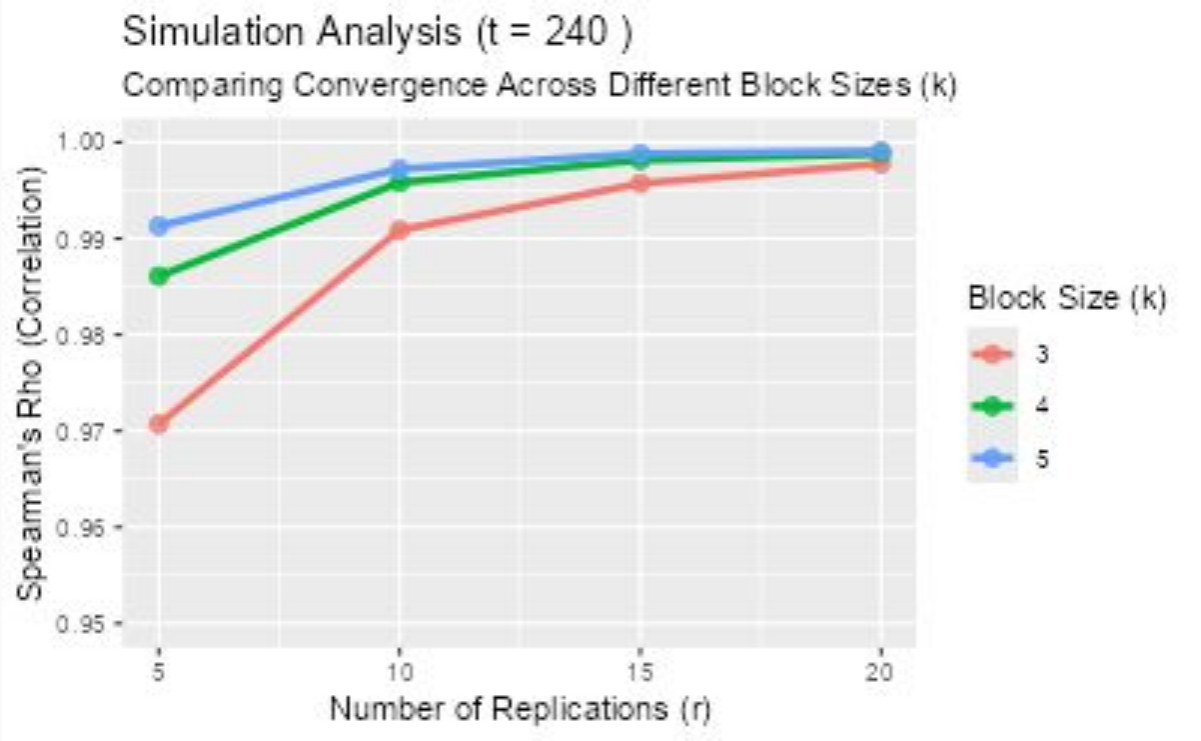
- Model Robustness Check (Sensitivity Analysis) -> Run the BT model using the "raw data including all respondents," and then run the model again using the "clean data after removing questionable responses. If the ρ value remains very high,it demonstrate high robustness and are not significantly influenced by extreme values

Graph

- [Github link]
- 

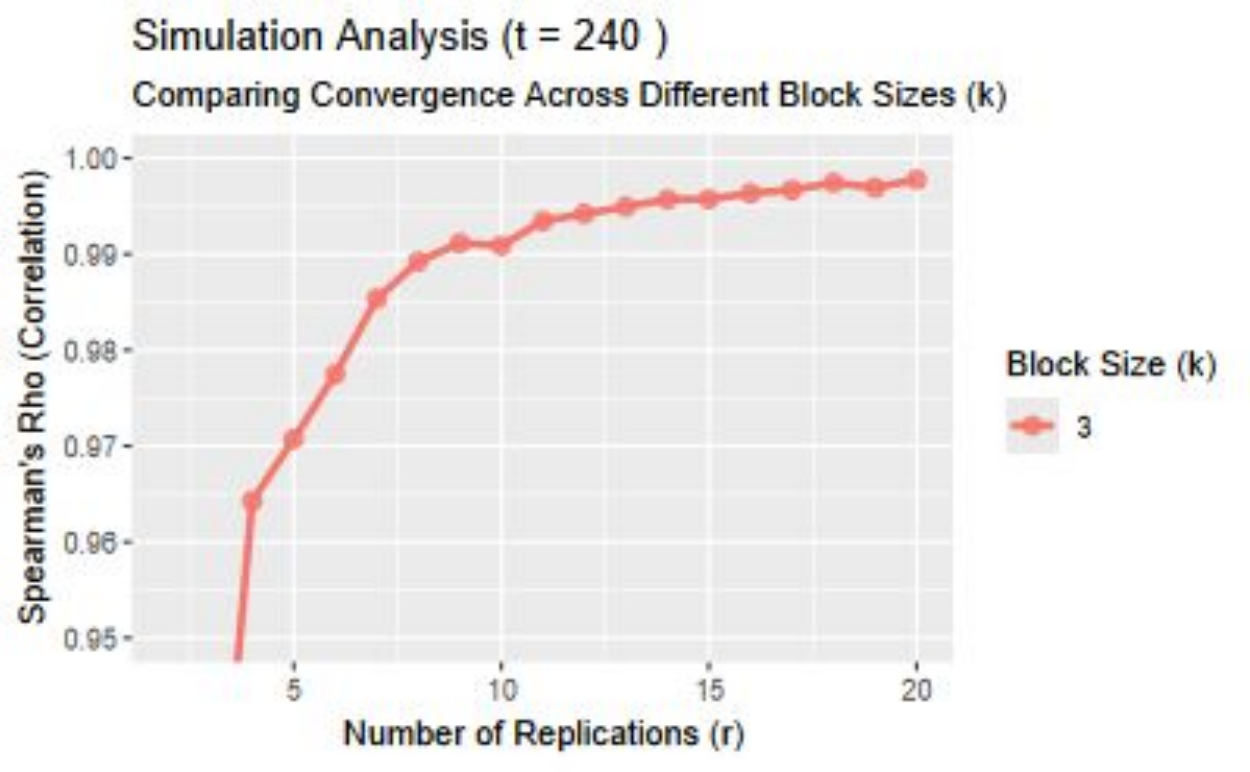
Link

Graph

- 
Simulation Analysis (t = 240)
Comparing Convergence Across Different Block Sizes (k)
Spearman's Rho (Correlation)
1.00
0.99
0.98
0.97
0.96
0.95
5
10
15
20
Number of Replications (r)
Block Size (k)
3
4
5

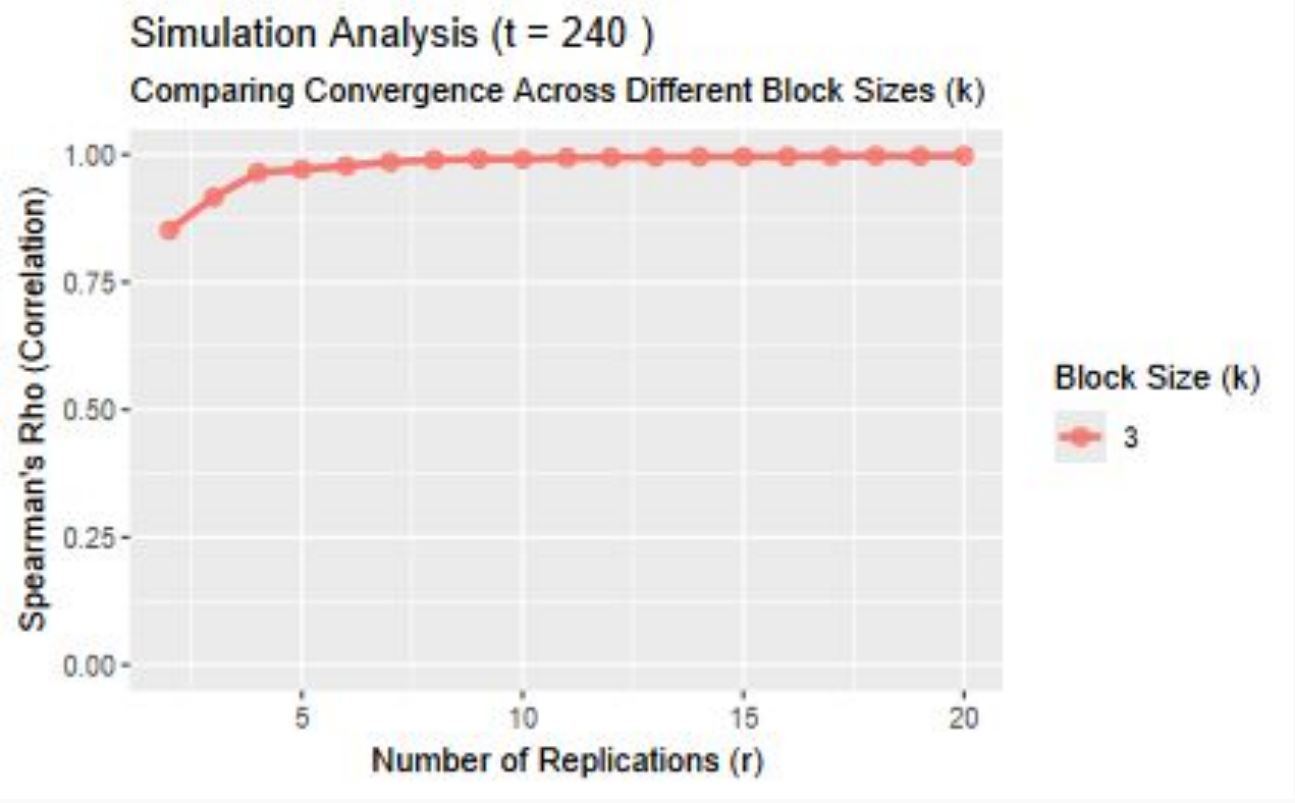
Link

Graph

- 
Simulation Analysis (t = 240)
Comparing Convergence Across Different Block Sizes (k)
Spearman's Rho (Correlation)
1.00
0.99
0.98
0.97
0.96
0.95
5
10
15
20
Number of Replications (r)
Block Size (k)
3

Link

Graph

- 
Simulation Analysis (t = 240)
Comparing Convergence Across Different Block Sizes (k)
Spearman's Rho (Correlation)
1.00
0.75
0.50
0.25
0.00
5
10
15
20
Number of Replications (r)
Block Size (k)
3

2026-03-24

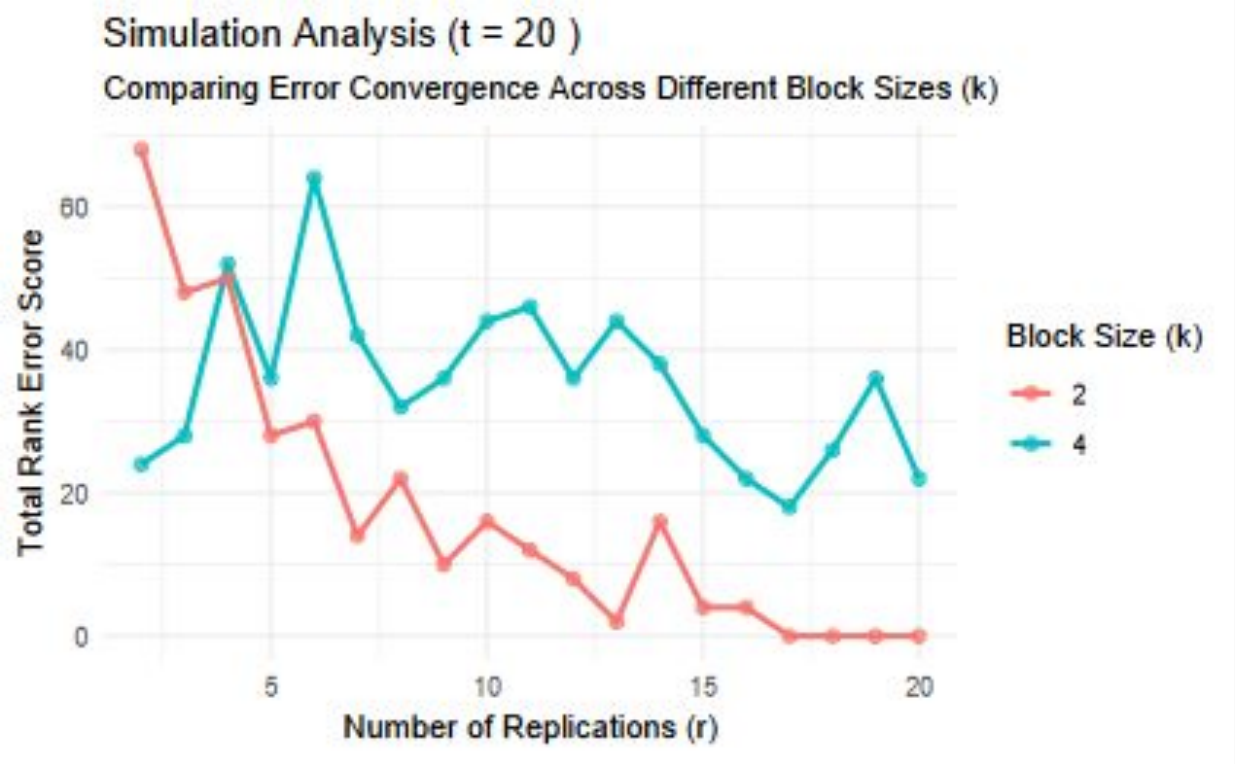
what is 'l' in alpha design

- Purpose of this parameter might can be explained by:
  "field trials with large number of crop varieties cannot always be laid out in a single location or in a single season. Therefore, it is desired that variation due to location or time periods may also be controlled along with controlling variation within location or within time period."
  Parsad, R., Gupta, V. K., Batra, P. K., Satpati, S. K., & Biswas, P. (2007). *Monograph on α-designs*. Indian Agricultural Statistics Research Institute (ICAR).
- So **I think** we can use it as a way to mark how many groups of respondents we have, for example business travelers vs leisure travelers.
- For non agriculture situation, maybe season for time series analysis, different religions for political support analysis in Malaysia

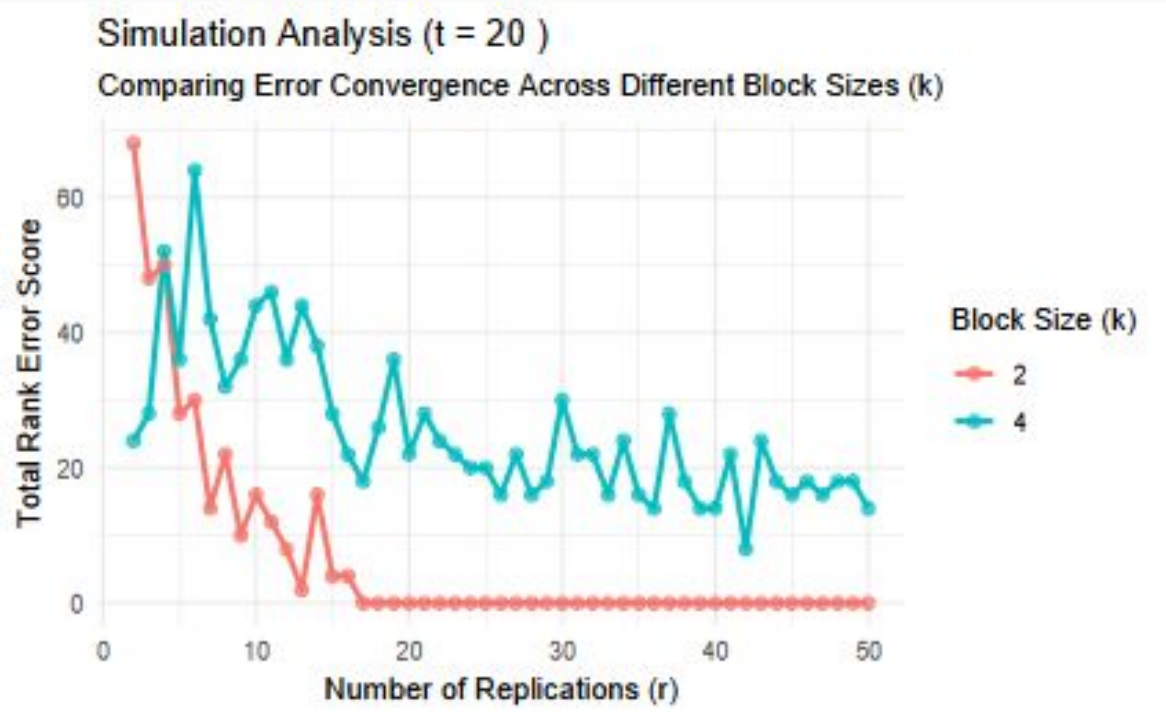
L vs r

- r(Replication):Repeating the same question within the same group.
- L(Location):Separating the experiment into completely different environments / batches / kinds of people / time.

Error when k = 2 & 4

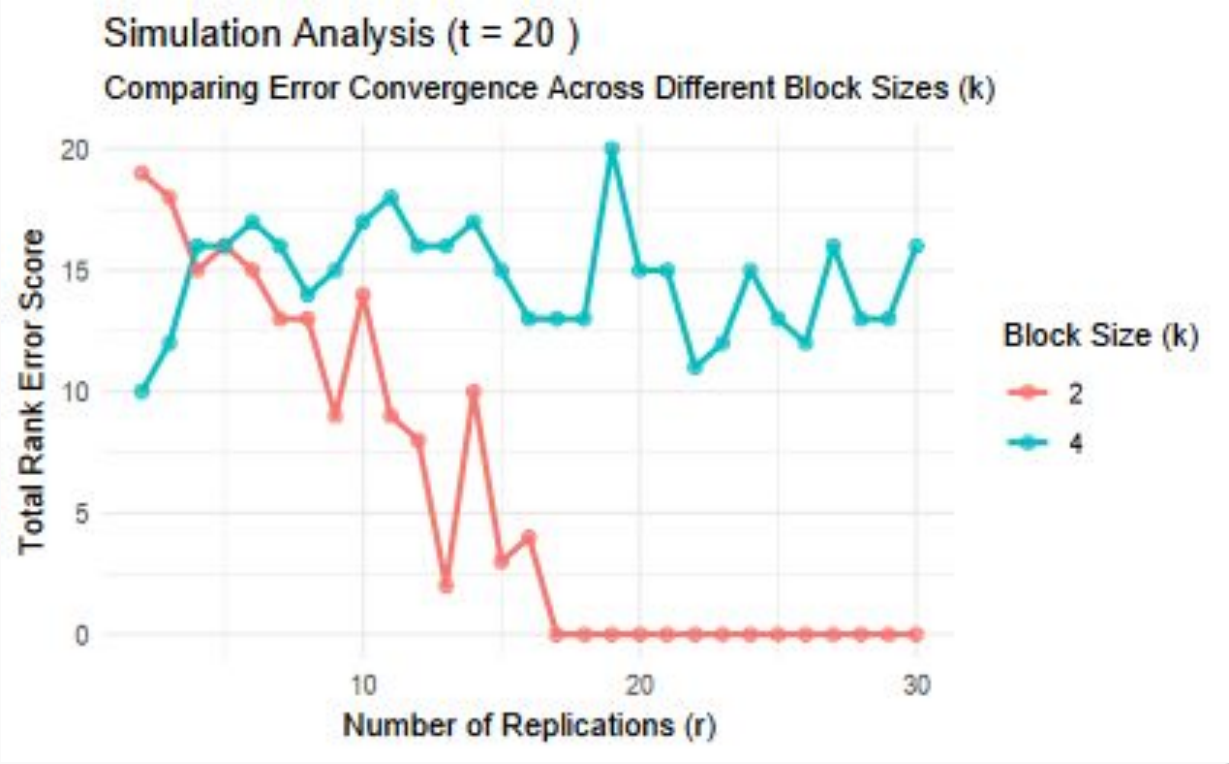


Error when k = 2 & 4

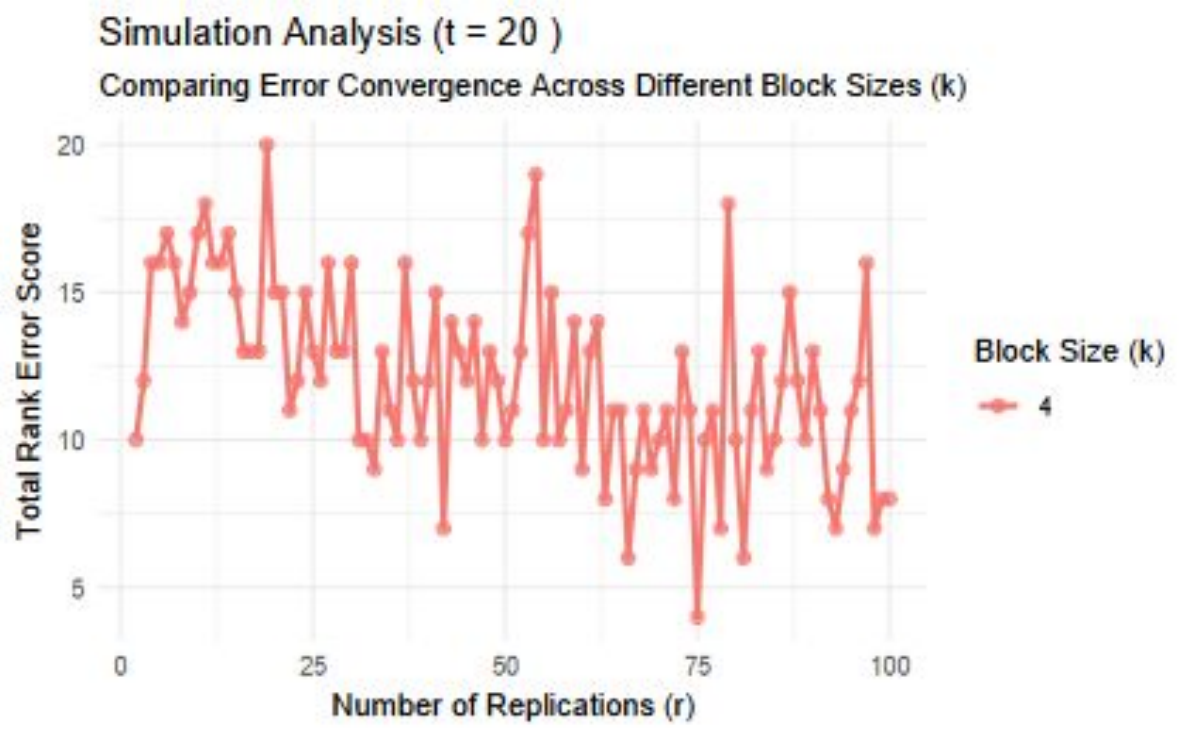


Error when k = 2 & 4, r =30

*show how many treatment not in right place



Error when k = 4, r =100



The code

Link

2026-03-17

what is 'l' in alpha design

- The document, shows that l = location
  But it didn't shows why we need location for this function.
  From the description, "FielDHub is an R Shiny design of experiments (DOE) app that aids in the creation of traditional, unreplicated, augmented and partially replicated (p-rep) designs **applied to agriculture, plant breeding, forestry, animal and biological sciences**."
  So *maybe* that's why we have l.

  ** need to generalize to non agriculture situation
  ** for our context , what did it do for us ?

definition of 'connection'

- “A design is ***disconnected*** **if you can separate the treatments into two groups, with no treatment from the first group ever appearing in the same block with a treatment from the second group**. A connected design is one that is not disconnected. In a connected design you can estimate all treatment differences. You cannot estimate all treatment differences in a disconnected design; in particular, you cannot estimate differences between treatments in different groups. Connectedness is obviously a very desirable property.”

Oehlert, G. W. (2010). *A first course in design and analysis of experiments*. Gary W. Oehlert.

Clarification

- show alphalattice result of simple example (g=20, k=4, m=g/k=5, r=2, etc.)
- Not like I expected
- Unlike traditional lookup tables that offer limited predefined configurations, FieIDHub generates alpha-lattice designs algorithmically.

```
> simple_alpha$fieldBook
```

	ID	LOCATION	PLOT	REP	IBLOCK	UNIT	ENTRY	TREATMENT
1	1	SITE1	101	1	1	1	10	G-10
2	2	SITE1	102	1	1	2	4	G-4
3	3	SITE1	103	1	1	3	1	G-1
4	4	SITE1	104	1	1	4	5	G-5
5	5	SITE1	105	1	2	1	3	G-3
6	6	SITE1	106	1	2	2	16	G-16
7	7	SITE1	107	1	2	3	12	G-12
8	8	SITE1	108	1	2	4	9	G-9
9	9	SITE1	109	1	3	1	17	G-17
10	10	SITE1	110	1	3	2	2	G-2
11	11	SITE1	111	1	3	3	18	G-18
12	12	SITE1	112	1	3	4	13	G-13
13	13	SITE1	113	1	4	1	6	G-6
14	14	SITE1	114	1	4	2	8	G-8
15	15	SITE1	115	1	4	3	15	G-15
16	16	SITE1	116	1	4	4	20	G-20
17	17	SITE1	117	1	5	1	7	G-7
18	18	SITE1	118	1	5	2	19	G-19
19	19	SITE1	119	1	5	3	11	G-11
20	20	SITE1	120	1	5	4	14	G-14

simulation

[File]

```
> head(final_results_df)
  Match_ID REP IBLOCK Winner Loser
1        1   1      1     15    18
2        2   1      2      6    16
3        3   1      3     12    20
4        4   1      4      4    17
5        5   1      5      2    11
6        6   1      6      3    19
>
```

Algorithm and test

[Algorithm]

[Test]

```
> print(sort(bt_abilities[, "ability"], decreasing = TRUE))
      G-1        G-6        G-5        G-4       G-15       G-16        G-7       G-13        G-3        G-9        G-2       G-12
24.814426  22.057256  16.542968   2.757169   0.000000  -2.757170  -5.514341  -8.271458 -11.028681 -13.785798 -16.542968 -19.300138
      G-8       G-18       G-17       G-19       G-11       G-14       G-10       G-20
22.057256 -24.814426 -27.571596 -35.843107 -41.357394 -44.114564 -46.871682 -49.628852
```

2026-03-10

Our challenge

- 200 data pre-collected, and we want to rank it
- Find out the preference toward the “solution” they gave, after we do the grouping (using method like alpha design/ RBIBD), we use pairwise comparison or BWS to get a brief knowledge of which is better than which
- After collecting the result we can use BT to calculate their “hidden power value”, finally rank them using it.

BT review

- BT:
  - BT model calculates **hidden power values** p_i for ranking items. Where Prob. i win j is calculated by $P(i>j) = p_i/(p_i+p_j)$
  - Model parameters are estimated using Maximum Likelihood Estimation (**MLE**) on pairwise data.
  - It works efficiently with incomplete comparisons, provided the comparison graph is **connected**.
- While the Elo system is a model designed for *dynamic, updating* based on the order of matches, the BT model treats all comparisons as a single dataset, making it *independent of time sequence*.

The grouping system(experiment design)

- We have a huge amount of data and want to make sure there are some connection between every data. Comparing every single data (with each other) will be too resource consuming.
- Thus we need a experiment design to make sure all the data somehow have <u>connection</u> with each other.

RIBD and Alpha design

- BIBD vs RIBD
  BIBD emphasize on equal λ (times 2 treatments are paired w/e.o.)
  RIBD emphasize on the completeness of data in one replication(r)
- We can use RBIBD to get a result with strong statistical relationship between the treatments (g) by dividing all the data into b blocks with size k. But again, RBIBD requires all the pair have the same λ, that is every pair of treatments must appear together in the same block exactly the same number of times.
- For RBIBD, It requires both properties of BIBD and RIBD
- So, we choose RIBD. But there is a problem with it.
- It didn’t regulate how we arrange the group, we might meet a situation that a pair meet too many time with each other when we randomized it.

RIBD and Alpha design

- So, we can choose alpha design, a method that use a <u>generating array</u> to decide how we divide the treatments into groups
- It requires that the number of treatments be a multiple of the block size $g = mk$
- For example, we have $g = 20$, we want $k = 4$, $r = 2$, so total comparisons is 40, when we have $k = 4$, then we will have $b = 10$ and m is defined as the number of blocks within each replication. In this case, we have $m = g/k = 5$.
- Next, we go and find generating array for $m = 5$, $r = 2$ and $k = 4$

RIBD and Alpha design

- Generating array for m = 5, r = 2 (columns) and k = 4 (rows)

1	1
1	2
1	3
1	4

RIBD and Alpha design

- Take each column of the generating array and does cyclic substitution with 1,2,....m, from 1st column of the generating array, we have

1	2	3	4	5
1	2	3	4	5
1	2	3	4	5
1	2	3	4	5

- Now, 2nd column of generating array

1	2	3	4	5
2	3	4	5	1
3	4	5	1	2
4	5	1	2	3

(GPS: this results in a *shifting*, which will finally create good connections)

RIBD and Alpha design

- Finally, we take the intermediate array and add m=5 to the second row (and multiples of m in following rows)

1	2	3	4	5
1 + 5	2 + 5	3 +5	4 +5	5 +5
1 +10	2 +10	3 +10	4 +10	5 +10
1 +15	2 +15	3 +15	4 +15	5 +15

- Now, 2nd column of generating array

1	2	3	4	5
2 +5	3 +5	4 +5	5 +5	1 +5
3 +10	4 +10	5 +10	1 +10	2 +10
4 +15	5 +15	1 +15	2 +15	3 +15

RIBD and Alpha design

- So we arrive at:

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20

- And for the 2nd column of generating array we arrive at:

1	2	3	4	5
7	8	9	10	6
13	14	15	11	12
19	20	16	17	18

Show package results of previous simple example

Example

```
alphalattice1 <- alpha_lattice(
t = 180,
k = 3,
r = 3,
l = 1,
plotNumber = 101,
locationNames = "group",
seed = 100
)
alphalattice1$infoDesign
head(alphalattice1$fieldBook, 10)
```

```
> head(alphalattice1$fieldBook, 10)
   ID LOCATION PLOT REP IBLOCK UNIT ENTRY TREATMENT
1   1    GROUP  101   1      1    1    50      G-50
2   2    GROUP  102   1      1    2   172     G-172
3   3    GROUP  103   1      1    3   178     G-178
4   4    GROUP  104   1      2    1    95      G-95
5   5    GROUP  105   1      2    2   164     G-164
6   6    GROUP  106   1      2    3    94      G-94
7   7    GROUP  107   1      3    1     8       G-8
8   8    GROUP  108   1      3    2   115     G-115
9   9    GROUP  109   1      3    3    90      G-90
10 10    GROUP  110   1      4    1   136     G-136
```

2026-02-23

Example of PBIBD

- Comparing treatment s - z, g=8 (no. of treatments), k=4 (block size), r=3 (no of times one treatment appears), b = no of blocks, k<g, bk=gr
- Why we want bk = gr？
- bk = gr = Total Number of Experimental Units
- cannot design an experiment with 3 blocks of size 4 if we have 5 treatments that we want to repeat 3 times

Revision

RBIBD (Resolvable-BIBD)

- Round Robin Tournament
- $g = 8, k = 2, r = 7, \lambda = 1$

Round (Set)	Block 1	Block 2	Block 3	Block 4
1	(1, 8)	(2, 7)	(3, 6)	(4, 5)
2	(2, 8)	(3, 1)	(4, 7)	(5, 6)
3	(3, 8)	(4, 2)	(5, 1)	(6, 7)
4	(4, 8)	(5, 3)	(6, 2)	(7, 1)
5	(5, 8)	(6, 4)	(7, 3)	(1, 2)
6	(6, 8)	(7, 5)	(1, 4)	(2, 3)
7	(7, 8)	(1, 6)	(2, 5)	(3, 4)

BIBD

- $g = 5, k = 2, r = 4, \lambda = 1$

Round 1	(1, 2)	(3, 4)	**Player 5**
Round 2	(1, 3)	(4, 5)	**Player 2**
Round 3	(1, 4)	(2, 5)	**Player 3**
Round 4	(1, 5)	(2, 3)	**Player 4**
Round 5	(2, 4)	(3, 5)	**Player 1**

Types of Group Divisible PBIBD

- Singular Group Divisible：Treatments in the same group are inseparable. If one treatment appears in a block, all its group-mates must also appear in that same block.
- Semi-Regular Group Divisible ：The design enforces strict fairness. Every block must contain an exactly equal number of treatments from each group.
- Regular Group Divisible：The general case of group divisible designs. Treatments are grouped, but the design does not impose the strict "inseparable" or "equal distribution" rules found in S-type or SR-type.

Types of Group Divisible PBIBD

- 8 Treatment, 4 groups:
  - Group 1: {1, 2}, Group 2: {3, 4}, Group 3: {5, 6}, Group 4: {7, 8}
- ~~Singular~~
  - must take the entire group or none of it
  - Valid Block: [1, 2, 3, 4]
  - Invalid Block: [1, 3, 5, 7]
- ~~Semi-Regular~~
  - must take the exact same number of treatments from every group.
  - Valid Block: [1, 3, 5, 7]
  - Invalid Block: [1, 2, 3, 4]

Types of Group Divisible PBIBD

- 8 Treatment, 4 groups:
  - Group 1: {1, 2}, Group 2: {3, 4}, Group 3: {5, 6}, Group 4: {7, 8}
- Regular
  - allows for uneven contributions from groups, unlike the strict rules of the previous
  - Valid Block: [1, 2, 3, 5]

Types of Group Divisible PBIBD

- Suitable for us？
- Singular：No, need the whole group when comparing, overemphasizing within-group comparisons
- Semi-Regular：Too Strict. Hard to design.
- Regular：better

Other Types of PBIBD

- Triangular：Relationships are defined by conceptually arranging treatments in a triangular or matrix pattern (specifically, the positions above the diagonal of a square matrix).
- Latin Square type : Relationships are determined by arranging treatments in a square grid. Two treatments are related if they share the same row or column (similar to a Sudoku structure).
- Cyclic : Relationships are determined by the numerical difference between treatment labels. Two treatments are considered related (associates) if the difference between their numbers belongs to a specific, fixed set of values.

Other Types of PBIBD - Triangular

- Triangular
  ：Requirement:
  g = n(n-1)/2,
  we use g = 10 here

  *But for our 400 treatment, it didn't suit n(n-1)/2

	Col 1	Col 2	Col 3	Col 4	Col 5
Row 1	x	**1**	**2**	**3**	**4**
Row 2	1	x	**5**	**6**	**7**
Row 3	2	5	x	**8**	**9**
Row 4	3	6	8	x	**10**
Row 5	4	7	9	10	x

匯出到試算表

The Rules:

- **Associates (Friends):** Two treatments are "friends" if they lie in the **same row** or the **same column**.
- **Example for Treatment 1:**
  - It is in **Row 1** and **Column 2**.
  - **Friends (Same Row):** 2, 3, 4
  - **Friends (Same Col):** 5, 6, 7 (See the symmetric positions)
  - **Strangers:** 8, 9, 10 (They share no row or column with 1).

Other Types of PBIBD - Triangular

	Col 1	Col 2	Col 3	Col 4	Col 5
Row 1	x	**1**	**2**	**3**	**4**
Row 2	1	x	**5**	**6**	**7**
Row 3	2	5	x	**8**	**9**
Row 4	3	6	8	x	**10**
Row 5	4	7	9	10	x

Other Types of PBIBD - Latin Square

- Latin Square type : g is a perfect square, for g = 9

	Col 1	Col 2	Col 3
Row 1	**1**	**2**	**3**
Row 2	**4**	**5**	**6**
Row 3	**7**	**8**	**9**

匯出到試算表

The Rules:

- **Associates (Friends):** Two treatments are friends if they share the **same row** OR the **same column**.
- **Example for Treatment 1:**
  - **Row Friends:** 2, 3
  - **Column Friends:** 4, 7
  - **Strangers:** 5, 6, 8, 9 (Diagonal positions).

Other Types of PBIBD - Cyclic

- Cyclic :Arrange treatments 1 to 8 in a circle

  Long path length, weak statistical relation between front and the middle treatment.

The Logic: Arrange treatments 1 to 8 in a circle (like a clock face). We define "friendship" based on **distance**.

Rule: Let's say "Friends" are those with a **difference (distance) of 1**.

The Layout: `(1) – (2) – (3) – (4) – (5) – (6) – (7) – (8) – (1)...`

The Rules:

- **Example for Treatment 1:**
  - **Friends (Distance 1):** 2 and 8 (8 is connected to 1 in a cycle).
  - **Strangers:** 3, 4, 5, 6, 7 (Too far away).
- **Example for Treatment 4:**
  - **Friends:** 3 and 5.
  - **Strangers:** Everyone else.

Alpha designs and RIBD -- which is feasible/better?

Experimental Design: Why Alpha Design?

- The Constraint:
  - High Item Count N = 200 .
  - Small Block Size k = 5 for BWS.
- Why NOT RIBD:
  - **Infeasible:** Combinatorial explosion makes λ=1 impossible.
  - **Resource Heavy:** Requires unrealistic sample sizes.
- Why Alpha Design:
  - **Feasible:** Optimized for large N.
  - **Efficient:** Uses **Partial Balance** to reduce survey complexity.

Link

FieIDHub alpha design

- FieIDHub

2026-01-05

A first course in design and analysis of experiments.

Oehlert, G. W. (2010).
***A first course in design and analysis of experiments*.**
Gary W. Oehlert.

Example of PBIBD

- Comparing treatment s - z, g=8 (no. of treatments), k=4 (block size), r=3 (no of times one treatment appears), k<g, bk=gr, ~~for now, r = 1~~

Block ID	Treatments			
1	s	w	t	x
2	u	y	v	z
3	s	w	u	y
4	t	x	v	z
5	s	w	v	z
6	t	x	u	y

Example of PBIBD

Obs A: Frequent Encounters

- Look at Pair (s, w).(in red)
- They appear together in Blocks 1, 3, and 5.
- Total Co-occurrence: 3 times.

Obs B: Occasional Encounters

- Look at Pair (s, t).(in blue)
- They appear together only in Block 1.
- Total Co-occurrence: 1 time.

Block ID	Treatments			
1	s	w	t	x
2	u	y	v	z
3	s	w	u	y
4	t	x	v	z
5	s	w	v	z
6	t	x	u	y

Unlike a BIBD, not all pairs meet with the same frequency. Relationships are unequal.

12 | 1 2 3 4 7 9 12

We see, for example, that treatment 1 occurs three times with treatments 5 and 9, and four times with all other treatments.

The design rules for a PBIBD are fairly complicated:

Requirements for PBIBD

1. There are g treatments, each used r times. There are b blocks of size $k < g$. Of course, $bk = gr$. No treatment occurs more than once in a block.
2. There are m *associate classes*. Any pair of treatments that are ith associates appears together in λ_i blocks. We usually arrange the λ_i values in decreasing order, so that first associates appear together most frequently.

   Associate classes
3. All treatments have the same number of ith associates, namely ρ_i.

   ρ_i ith associates
4. Let A and B be two treatments that are ith associates, and let p^i_{jk} be the number of treatments that are jth associates of A and kth associates of B. This number p^i_{jk} does not depend on the pair of ith associates chosen. In particular, $p^i_{jk} = p^i_{kj}$.

The PBIBD is partially balanced, because the variance of $\widehat{\alpha}_i - \widehat{\alpha}_j$ depends upon whether i, j are first, second, or mth associates. The randomization of a PBIBD is just like that for a BIBD.

Randomize PBIBD like BIBD

Let's check the design given above and verify that it is a PBIBD. First note that $g = 12$, $k = 7$, $r = 7$, $b = 12$, and no treatment appears twice in a block. Next, there are two associate classes, with first associates appearing together four times and second associates appearing together three times. The pairs (1,5), (1,9), (2,6), (2,10), (3,7), (3,11), (4,8), (4,12), (5,9), (6,10), (7,11), and (8,12) are second associates; all other pairs are first associates. Each treatment has nine first associates and two second associates. For any pair of

Example of PBIBD

- The properties
  - **Associate Classes**: Pairs are classified into m classes. i^{th} associates appear together λ_i times.
    - (s,w) meet each other 3 times, but individually, they pair with others 1 time; the same is true for (u,y) and so on
  - **Uniformity**: All treatments have the same number of ith associates n_i.
    - Same as above, but emphasis on “every one have to be the same”, as when comparing, for any treatment, it only meet 1 treatment 3 times, and others 1 time.
  - **Structural Symmetry**: No matter which treatment you look at, the "web of connections" looks identical. This ensures the statistics are valid.

Example of RBIBD (Resolvable-BIBD)

- Round Robin Tournament
- $g = 8$, $k = 2$, $r = 7$, $\lambda = 1$

Round (Set)	Block 1	Block 2	Block 3	Block 4
1	(1, 8)	(2, 7)	(3, 6)	(4, 5)
2	(2, 8)	(3, 1)	(4, 7)	(5, 6)
3	(3, 8)	(4, 2)	(5, 1)	(6, 7)
4	(4, 8)	(5, 3)	(6, 2)	(7, 1)
5	(5, 8)	(6, 4)	(7, 3)	(1, 2)
6	(6, 8)	(7, 5)	(1, 4)	(2, 3)
7	(7, 8)	(1, 6)	(2, 5)	(3, 4)

- r groups, with each group representing a complete set of treatments.
- all treatments appear exactly once.

Types of PBIBD

- Singular Group Divisible： Treatments in the same group are inseparable. If one treatment appears in a block, all its group-mates must also appear in that same block.
- Semi-Regular Group Divisible ：The design enforces strict fairness. Every block must contain an exactly equal number of treatments from each group.
- Regular Group Divisible：The general case of group divisible designs. Treatments are grouped, but the design does not impose the strict "inseparable" or "equal distribution" rules found in S-type or SR-type.

Types of PBIBD

- 8 Treatment, 4 groups:
  - Group 1: {1, 2}, Group 2: {3, 4}, Group 3: {5, 6}, Group 4: {7, 8}
- Singular
  - Valid Block: [1, 2, 3, 4]
  - Invalid Block: [1, 3, 5, 7]
  - must take the entire group or none of it
- Semi-Regular
  - Valid Block: [1, 3, 5, 7]
  - Invalid Block: [1, 2, 3, 4]
  - must take the exact same number of treatments from every group.

Types of PBIBD

- 8 Treatment, 4 groups:
  - Group 1: {1, 2}, Group 2: {3, 4}, Group 3: {5, 6}, Group 4: {7, 8}
- Regular
  - Valid Block: [1, 2, 3, 5]
  - allows for uneven contributions from groups, unlike the strict rules of the previous

Types of PBIBD

- Suitable for us？
- Singular：No, need the whole group when comparing, overemphasizing within-group comparisons
- Semi-Regular：Too Strict. Hard to design.
- Regular：better

Types of PBIBD

- Triangular：Relationships are defined by conceptually arranging treatments in a triangular or matrix pattern (specifically, the positions above the diagonal of a square matrix).
- Latin Square type : Relationships are determined by arranging treatments in a square grid. Two treatments are related if they share the same row or column (similar to a Sudoku structure).
- Cyclic : Relationships are determined by the numerical difference between treatment labels. Two treatments are considered related (associates) if the difference between their numbers belongs to a specific, fixed set of values.

Types of PBIBD

- Triangular
  ：Requirement:
  g = n(n-1)/2,
  we use g = 10 here

  *But for our 400 treatment, it didn't suit n(n-1)/2

	Col 1	Col 2	Col 3	Col 4	Col 5
Row 1	x	**1**	**2**	**3**	**4**
Row 2	1	x	**5**	**6**	**7**
Row 3	2	5	x	**8**	**9**
Row 4	3	6	8	x	**10**
Row 5	4	7	9	10	x

匯出到試算表

The Rules:

- **Associates (Friends):** Two treatments are "friends" if they lie in the **same row** or the **same column**.
- **Example for Treatment 1:**
  - It is in **Row 1** and **Column 2**.
  - **Friends (Same Row):** 2, 3, 4
  - **Friends (Same Col):** 5, 6, 7 (See the symmetric positions)
  - **Strangers:** 8, 9, 10 (They share no row or column with 1).

Types of PBIBD

- Latin Square type : g is a perfect square, for g = 9

	Col 1	Col 2	Col 3
Row 1	**1**	**2**	**3**
Row 2	**4**	**5**	**6**
Row 3	**7**	**8**	**9**

匯出到試算表

The Rules:

- **Associates (Friends):** Two treatments are friends if they share the **same row** OR the **same column**.
- **Example for Treatment 1:**
  - **Row Friends:** 2, 3
  - **Column Friends:** 4, 7
  - **Strangers:** 5, 6, 8, 9 (Diagonal positions).

Types of PBIBD

- Cyclic :Arrange treatments 1 to 8 in a circle

  Long path length, weak statistical relation between front and the middle treatment.

The Logic: Arrange treatments 1 to 8 in a circle (like a clock face). We define "friendship" based on **distance**.

Rule: Let's say "Friends" are those with a **difference (distance) of 1**.

The Layout: `(1) – (2) – (3) – (4) – (5) – (6) – (7) – (8) – (1)...`

The Rules:

- **Example for Treatment 1:**
  - **Friends (Distance 1):** 2 and 8 (8 is connected to 1 in a cycle).
  - **Strangers:** 3, 4, 5, 6, 7 (Too far away).
- **Example for Treatment 4:**
  - **Friends:** 3 and 5.
  - **Strangers:** Everyone else.

Example of RBIBD

- Square Lattice Design：Treatments are arranged conceptually in a perfect square grid. Blocks are formed by taking an entire Row (in the first replication) or an entire Column (in the second replication).

  Rigid constraint where the total number of treatments must be a perfect square. So using this, our block size will have to be k = 20. It is too big!

Example of RBIBD

- **The Setup: The Grid** Arrangement of Treatments 1 to 9:

$$\begin{matrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{matrix}$$

Square Lattice Example

- **Logic:** "Straight Lines Only." You can only group treatments that form a straight horizontal line (Row) or a straight vertical line (Column).
- **Valid Block:** `[ 1, 2, 3 ]`
  - **Why?** This is **Row 1**. It is a complete straight line in the grid.
  - *Also Valid:* `[ 1, 4, 7 ]` (This is **Column 1**).
- **Invalid Block:** `[ 1, 2, 4 ]`
  - **Why?** This forms an **"L" shape** (1 and 2 are Row 1, but 4 is Row 2). You cannot turn corners; you must stick to one straight line.

Package ‘agricolae’

Felipe de Mendiburu(2025).
Package ‘agricolae’ **p34**

Alpha design example

- Rmd
- Not found, BT model use with alpha design

2025-12-01

A first course in design and analysis of experiments.

Oehlert, G. W. (2010).
***A first course in design and analysis of experiments*.**
Gary W. Oehlert.

Why Choose RIBD over BIBD?

- Operational Flexibility
  - The experiment is modular (grouped into complete replicates).
  - Can be run in distinct batches
  - Risk Management: If resources run out, entire replications can be dropped without ruining the whole experiment.
- Feasibility for Large Scale
  - For large number of treatments (e.g., g=400), a perfect BIBD requires an impractical number of blocks/samples
  - RIBD allows for valid statistical analysis with a manageable sample size.

Explain and example

Why Choose RIBD over BIBD?

- Hierarchical Error Control
  - **Blocks** control local variation; **Replicates** control global variation.
  - Allows the researcher to account for large-scale shifts (e.g., Week 1 vs. Week 2 effects) which non-resolvable BIBDs cannot easily separate.

Example: Imagine you run an experiment over two weeks.

Week 1 (Replicate 1): The weather was sunny.

Week 2 (Replicate 2): The weather was rainy.

The Benefit: Because Replicate 1 and Replicate 2 are separate complete sets, the math can easily calculate and remove the "Weather Effect" (Week 1 vs. Week 2) from the results. A non-resolvable BIBD mixes everything up, making it very hard to separate these large-scale environmental shifts.

RIBD (**Resolvable** incomplete block design)

- A resolvable design is one in which the blocks can be arranged into r groups, with each group representing a complete set of treatments.
- Example, comparing A,B,C,D
- Total no. of treatment = 4, we set block size = 2
- Rep 1
  - Block 1 : A vs B
  - Block 2 : C vs D
- Rep 2
  - Block 1 : A vs C
  - Block 2 : B vs D

RIBD (Resolvable incomplete block design)

- Rep 1
  - Block 1 : A vs B
  - Block 2 : C vs D
- Rep 2
  - Block 1 : A vs C
  - Block 2 : B vs D
- Now we have $\lambda_{AB} = \lambda_{CD} = 1$ in rep 1, but $\lambda_{AC}, \lambda_{AD}, \lambda_{BC}, \lambda_{BD} = 0$
- Similarly rep 2, we have $\lambda_{AC} = \lambda_{BD} = 1$, others = 0

RIBD (Resolvable incomplete block design)

- In each Rep, all treatments (A, B, C, D) appear exactly once.
- This forms a Complete Replicate (Resolvable).
- Contrast: An Incomplete Replicate would fail to cover all treatments (e.g., Block 1: {A, B}, Block 2: {A, D} — C is missing).

PBIBD (Partially Balanced Incomplete Block Design)

- Solving BIBD huge comparison requirement.
- 1st we need to go and find a table (like in alpha design)
- So how they design the table ?
  - The sample size is consistent bk = gr
  - Co-occurrence frequencies λ_i are fixed for each associates
  - Every treatment has the exact same number of partners p_i in each associates
  - The intersection of relationships p^i_{jk} is constant and symmetric.

PBIBD (Partially Balanced Incomplete Block Design)

- ref

2025-11-24

Design of Experiments

A first course in design and analysis of experiments.

Oehlert, G. W. (2010).
***A first course in design and analysis of experiments*.**
Gary W. Oehlert.

Parameter for BIBD

Parameter	Meaning	In Our Study
v / g / t	Total number of items/treatments	~400
k	Block size, how many treatments in the block ** k < g	
b	Total number of blocks or , simply said , total no of respondent	
r	how many times one treatment appears	
λ (lambda)	How many times a pair appears together	

Concept

BIBD: Core Concepts

1. Incomplete Block:

Definition: A design where the block size (k) is smaller than the total number of treatments (g).

Why:

- Used when a single block cannot physically accommodate all treatments.
- Including too many treatments causes subject fatigue or inaccurate responses.

2. Balanced:

Definition: The number of blocks in which a specific pair of treatments appear together.

Benefit: For a design to be a BIBD, the pairwise occurrence λ must be identical for every single possible pair of treatments.This symmetry ensures that all pairwise comparisons (Treatment A vs. B) have the same variance/precision.

Two Golden Rules **N = bk = rg**

- Total Count N = bk = rg

  To ensure the Total Response Volume (answers provided by all respondents) matches the Total Data Requirement (ratings needed for all items).

- Example
  We need 50 ratings r for each of the 10 products g. Total = 500 ratings.

  Each respondent b only rates 5 products k.
  bk = rg

  We must recruit exactly 100 respondents 100 * 5 = 10 * 50 to satisfy the demand.

Two Golden Rules $\lambda(g - 1) = r(k - 1)$ ——part 1

- The Balancing $\lambda(g - 1) = r(k - 1)$
  To ensure every pair of items is compared directly by the same number of respondents.
  Example:
  Consider a specific treatment, Product X, within the set of g total treatments.

  $r(k - 1)$:Product X appears in exactly r blocks (questionnaires). In each of these blocks, there are $(k - 1)$ other experimental units available for other treatments. Consequently, Product X co-occurs with other treatments a total of $r(k - 1)$ times across the entire experiment.

Two Golden Rules $\lambda(g - 1) = r(k - 1)$ ——part 2

- The Balancing $\lambda(g - 1) = r(k - 1)$
  To ensure every pair of items is compared directly by the same number of respondents.
  $\lambda(g - 1)$:There are $(g - 1)$ other distinct treatments in the experiment (excluding Product X). For the design to be balanced, Product X must co-occur with each of these $(g - 1)$ treatments exactly λ times. Thus, the sum of co-occurrences required to pair Product X with every rival is $\lambda(g - 1)$.

  Conclusion: By equating the total co-occurrence slots to the required balanced distribution, we derive the identity $\lambda(g - 1) = r(k - 1)$.

Link

Resolvable Incomplete Block Design

- A design where blocks can be grouped into separate sets called Replicates
- Each Replicate contains exactly one complete set of all treatments.
- For 500 treatments, Replicate 1 consists of 50 blocks that cover all 500 items once.
- Ideal for management. Replicates can be run at different times, locations, or dropped if resources fail.

Link

Alpha Designs

- A method to construct Resolvable Incomplete Block Designs (RIBD)
- When the number of treatments g is large and is a multiple of the block size k (i.e., g=mk)
- Uses a "Generating Array" and "Cyclic Expansion" to mathematically ensure that treatments are mixed thoroughly across the replicates.

Link

Alpha Designs

- Example
- Total Treatments g: 20
- Block Size k: 4 (4 treatments per block)
- Replicates r: 2 (We will create 2 full replications)
- Number of Groups m: g/k

  = 20/4

  = 5

Link

Alpha Designs

C.4 Alpha Designs

Alpha Designs are resolvable block designs for $g = mk$ treatments in $b = mr$ blocks of size k. These tables give the initial alpha arrays for $5 \le m \le 15$, block sizes from 4 up to the minimum of m and $100/m$, and up to four replications. These tables are adapted from Table 2 of Patterson, Williams, and Hunter (1978).

$m = 5$, $4 \le k \le 5$

1	1	1	1
1	2	5	3
1	3	4	5
1	4	3	2
1	5	2	4

$m = 6$, $4 \le k \le 6$

1	1	1	1
1	2	6	5
1	4	3	6
1	3	4	2
1	5	2	3
1	6	2	4

$m = 7$, $4 \le k \le 7$

1	1	1	1
1	2	4	3
1	3	7	5
1	5	6	2
1	4	3	7
1	6	2	4
1	7	5	6

$m = 8$, $4 \le k \le 8$

1	1	1	1
1	2	3	7
1	4	8	2
1	6	4	5
1	3	6	4
1	5	2	7
1	7	1	3
1	8	7	6

$m = 9$, $4 \le k \le 9$

1	1	1	1
1	2	9	8
1	4	7	5
1	8	3	4
1	3	4	6
1	5	2	7
1	6	8	3
1	7	6	2
1	9	5	8

$m = 10$, $4 \le k \le 10$

1	1	1	1
1	2	10	6
1	4	7	10
1	6	8	3
1	5	6	7
1	7	4	2
1	8	3	5
1	9	5	8
1	10	9	3
1	3	7	4

$m = 11$, $4 \le k \le 9$

1	1	1	1
1	2	7	8
1	5	9	2
1	10	8	6
1	3	4	7
1	6	2	4
1	7	6	11
1	4	10	5
1	8	5	2

$m = 12$, $4 \le k \le 8$

1	1	1	1
1	2	3	4
1	8	6	2
1	10	7	5
1	5	12	9
1	12	4	11
1	11	5	8
1	6	2	7

$m = 13$, $4 \le k \le 7$

1	1	1	1
1	2	5	11
1	4	9	12
1	10	3	2
1	13	11	7
1	9	6	13
1	7	8	9

$m = 14$, $4 \le k \le 7$

1	1	1	1
1	2	9	11
1	10	11	8
1	12	14	3
1	3	7	2
1	6	12	13
1	4	2	12

$m = 15$, $4 \le k \le 6$

1	1	1	1
1	2	9	8
1	4	13	15
1	8	3	6
1	11	14	12
1	15	4	9

Link

Alpha Designs

Step 2: Cyclic Expansion

Let's construct the **Second Replication (Rep 2)**. We use the seed column: $[1, 2, 3, 4]$. **Rule:** For each subsequent block, add **+1** to the numbers (cycling within the range 1 to 5).

	Block 1 (Seed)	Block 2 (+1)	Block 3 (+1)	Block 4 (+1)	Block 5 (+1)
Row 1	1	2	3	4	5
Row 2	2	3	4	5	1
Row 3	3	4	5	1	2
Row 4	4	5	1	2	3

EXPORT TO SHEETS

Link

Alpha Designs

Step 3: Adding Offsets (Finalizing IDs)

Now, convert these numbers (1-5) into actual Treatment IDs (1-20). **Offset Rule ($m = 5$):**

- Row 1: Add **0**
- Row 2: Add **5** (m)
- Row 3: Add **10** ($2m$)
- Row 4: Add **15** ($3m$).

Let's calculate the contents for **Block 1** and **Block 2**:

Calculation for Block 1:

- Row 1: $1 + 0 = \textbf{Treatment 1}$
- Row 2: $2 + 5 = \textbf{Treatment 7}$
- Row 3: $3 + 10 = \textbf{Treatment 13}$
- Row 4: $4 + 15 = \textbf{Treatment 19} \rightarrow$ **Block 1 = {1, 7, 13, 19}**

Link

Alpha Designs

Calculation for Block 2:

- Row 1: $2 + 0 = \textbf{Treatment 2}$
- Row 2: $3 + 5 = \textbf{Treatment 8}$
- Row 3: $4 + 10 = \textbf{Treatment 14}$
- Row 4: $5 + 15 = \textbf{Treatment 20} \rightarrow$ **Block 2 = {2, 8, 14, 20}**

Repeat this for all 5 blocks, and your design for Replication 2 is complete!

2025-11-17

Bradley Terry Scalable

Ella Kaye,David Firth
Bradley Terry Scalable
https://ellakaye.github.io/BradleyTerryScalable/

Link

PDF & RMD file

- PDF
- RMD

Best-worst scaling improves measurement of first impressions

Burton, N., Burton, M., Rigby, D. et al.
Best-worst scaling improves measurement of first impressions.
Cogn. Research 4, 36 (2019). https://doi.org/10.1186/s41235-019-0183-2

Abstract

Best-Worst Scaling (BWS) for Measuring Facial First Impressions

- The Challenge: Likert ratings are simple but have response issues and limited reliability for subjective impressions.
- The Alternative: Best-Worst Scaling (BWS), a forced-choice method, was examined as a potential high-reliability alternative.
- Group Level: BWS scores correlated almost perfectly with Likert scores, confirming they measure the same impressions.
- Individual Level (Superiority): BWS significantly outperforms Likert ratings, showing greater:
  - Test-Retest Reliability
  - Predictive Validity

Study 1: Validity (Attractiveness)

- Core Goal: Check group-level consistency between BWS and Likert, and test their validity (predictive power).
- BWS Method: 30 items (faces), 30 trials.(use Sawtooth Software's Lighthouse Studio)
  - 5 items per subset.
  - Select "Most Attractive" and "Least Attractive."
- Criterion Task: Rank 3 faces from most attractive to least attractive.
- Results: BWS scores were a more accurate predictor of participants' rankings in the criterion task (AICc difference > 10).

Study 2: Test-Retest Reliability (Distinctiveness)

- Core Goal: Directly compare the Test-Retest Reliability of BWS and Likert.
- Design: Two participant groups (BWS vs. Likert), tested twice with an interval of approximately 3-5 days.
- BWS Method: 30 items, 30 trials, 5 items per subset.
  - Select "Most Distinctive" and "Least Distinctive."
- Consistency Analysis: Calculated correlation (Pearson's r) between Session 1 and Session 2 scores.
- Results: BWS correlation (r = .66) was significantly higher than Likert (r = .53), showing greater BWS reliability.

Study 3: Reliability Replication (Trustworthiness)

- Core Goal: Replicate Study 2's high reliability findings and validate the feasibility of Open-Source Tools (R + Qualtrics).
- Materials: 31 "ambient" faces (from the US 10k database).
- BWS Design (BIBD): 31 items, 31 trials.
  - 6 items per subset.
  - Select "Most Trustworthy" and "Least Trustworthy."
- Results: BWS reliability (r = .76) was again significantly higher than Likert (r = .63).

Summary: Why is BWS Superior?

- Group Level: BWS and Likert scores were highly correlated ($r > .86$), confirming both measure the same impressions.
- Individual Level: BWS clearly outperformed Likert:
  - **Higher Validity (Study 1):** BWS more accurately predicted individual preferences in the ranking task.
  - **Higher Reliability (Studies 2 & 3):** BWS showed higher consistency in test-retest scores, indicating more stable participant responses.
- Conclusion: BWS reduced response bias and noise from participants, making it a more reliable tool for measuring individual differences.

2025-11-02

Clarification

What is the largest BT model we can have

- theoretically unlimited but is constrained primarily by computational power and data quality in practice

Clarification

What is the largest BT model we can have

- theoretically unlimited but is constrained primarily by computational power and data quality in practice

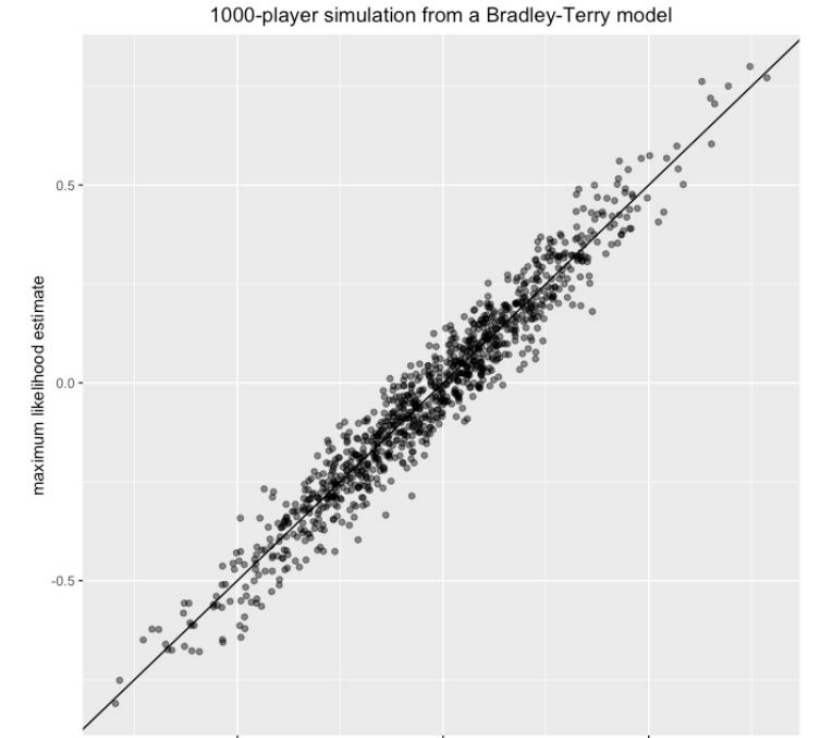
Clarification

how can we make a design with minimum number of comparisons?

- Will included in last slide, but not sure it is suitable to use as explanation or not

Huge data example

- As a bigger example, let's simulate a single instance of a fairly sparse tournament with 1000 items (or 'players'), and then fit the Bradley-Terry model to the resulting data:
- [BradleyTerryScalable]



Best-Worst Scaling: Theory, Methods and Applications.

Louviere, J. J., Flynn, T. N., & Marley, A. A. J. (2015).
Best-Worst Scaling: Theory, Methods and Applications.
Cambridge: Cambridge University Press.

BWS

- Best-Worst Scaling (BWS), also known as "BWS," is a choice modeling technique or discrete choice experiment (DCE) invented by Jordan Louviere in 1987.
- Steps:
  - Present Items: Respondents are shown a subset of items (e.g., brands, statements, features) taken from a master list.
  - Make Choices: They are asked to choose the single "best" (e.g., most important, most appealing, most preferred) item and the single "worst" (e.g., least important, least appealing, least preferred) item from that specific subset.

BWS

- Repeat Task: This task is repeated several times, with the items in the subset varying systematically according to a statistical design.

- The purpose of BWS is to measure subjective quantities (like preferences, attitudes, or perceived importance) on a measurement scale with known properties.
- Extension of the paired comparison method.
- Forces respondents to make trade-offs, provides more information than "best-only" tasks, and avoids many of the biases associated with traditional rating scales (like Likert scales).

Concept

Core Assumptions 1

1. The BWS Task and some definition

- BWS models are statistical models that treat choices as probabilities, not certainties.
  - "All BWS models are probabilistic discrete choice models."
- "Best" and "worst" are labels for two extremes of any scale (e.g., most/least important, heaviest/lightest).
  - "Here 'best' and 'worst' simply constitute a metaphor for any appropriate terms that define the extremes of a latent, subjective continuum."

Core Assumptions 2

2.Theoretical Framework Assumptions (RUT & FUT)

- Users must be able to pick the top and bottom items from a set of 3 or more.
  - "Best-worst scaling is based on the idea that a person faces choices among collections (“sets”) of three or more items or options, and can identify the best and the worst options in the collection."
- Assumes a person's underlying preference (utility) is somewhat random or unstable【RUT (Random Utility Theory)】.
  - "Random utility theory assumes that the decision process underlying the choices is deterministic, but the utilities have a random component."
- Assumes the preference is stable, but the act of choosing is random or imperfect.
  - ”Fixed utility theory”…

Core Assumptions 2

2.Theoretical Framework Assumptions (RUT & FUT) continued

- "The two theoretical perspectives can lead to choice models that are mathematically equivalent, even though they may imply potentially different economic or psychological processes (see Chapter 5)."

Core Assumptions 3

3.About Aggregating Data（related to our RESEARCH）

- The book's models are designed to analyze one person at a time.
- Trying to combine, or aggregate, data from many individuals to summarize a whole group is complex. This aggregation requires making "strong assumptions" which might be incorrect and can create significant problems for the analysis.

Core Paper

Preferences for tongue swab-based versus sputum-based testing in the context of TB care: a best-worst scaling exercise in Vietnam and Zambia.

Castro, M. D. M., Le, H., Muzazu, S., Pham, N., Trinh, T., Nyirenda, H. C., Shabalu, P., West, N., Phan, H., Cattamanchi, A., Denkinger, C. M., Muyoyeta, M., & Kerkhoff, A. D. (2025). Preferences for tongue swab-based versus sputum-based testing in the context of TB care: a best-worst scaling exercise in Vietnam and Zambia. **BMJ Global Health**, 10(10), e019092. https://doi.org/10.1136/bmjgh-2025-019092

Example of BWS design with BIBD

- Castro MDM, Le H, Muzazu S, Pham N, Trinh T, Nyirenda HC, et al. Preferences for tongue swab-based versus sputum-based testing in the context of TB care: a best-worst scaling exercise in Vietnam and Zambia. BMJ Global Health. 2025;10:e019092. https://doi.org/10.1136/bmjgh-2025-019092
- Used Best-Worst Scaling (BWS) to find out which of 16 features for TB testing were most important to patients and used a Balanced Incomplete Block Design (BIBD) to create the survey.
  - The total number of features (for other case, item / treatment / solution) being evaluated. v = 16
  - The number of features shown in each block (here, they present the block in the form of question). k =4
  - The total number of blocks (questions) each participant completed. b = 12
  - The number of replications (how many times each feature appeared for each participant). r = 3
  - The paper calls this a "near-BIBD," meaning this pairing might not be perfectly balanced for every single pair, so no λ（I think）

Example

Example of BWS design with BIBD

- [Supplementary File for Castro MdM, et al]
- Shows how to calculate minimum sample size.

$$N \geq \left(\frac{z_{\alpha/2}}{\delta}\right)^2 \frac{2.12n}{mt}.$$

Where:

- N = minimum sample size required to achieve the specified margin of error for estimated preference weights
- $Z_{\alpha/2}$ = 1.96 for 95% confidence
- δ = is the margin of error (MOE)
- n = 16 is the total number of features
- m = 4 is the number of features per task
- t = 12 is the number of tasks per respondent
- 2.12 = simulation-derived constant (Lipovetsky et al. 2015)

Using a margin of error δ = 0.143, the minimum sample size required per country is:

N ≥ (1.96 / 0.143) ^2 * (2.12 × 16) / (4 × 12) ≈ 133

Thus, a sample size of 133 participants per country is sufficient to achieve an approximate MOE of 0.143, ensuring acceptable precision in the estimation of utility scores derived from the BWS design. Figure 1 presents the estimated margin of error for a range of sample sizes using the current design.

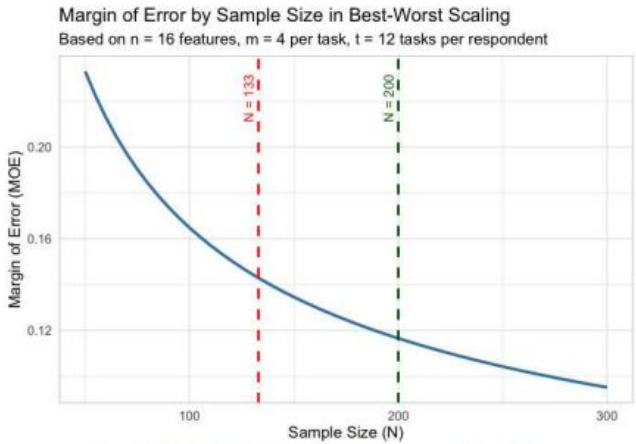


Figure 1. Margin of error by sample size estimations.

2025-10-27

RANK ANALYSIS OF INCOMPLETE BLOCK DESIGN

THE METHOD OF PAIRED COMPARISON

Bradley, R. A., & Terry, M. E. (1952). Rank Analysis of Incomplete Block Designs: I. The Method of Paired Comparisons. *Biometrika*, *39*(3/4), 324–345. https://doi.org/10.2307/2334029

Clarification

Group vs block vs repetition

- Group：a set of experiment repetitions, which often represents a distinct source of variation like different judges or different times. The primary statistical constraint is that the model's design requires these groups to be defined such that the true treatment ratings (π_{1u},..., π_{tu}) exist within each uth group, but are not necessarily identical from one group to the next.（**no. of judge not necessary = no. of group）

- ref：In certain cases, as noted above, repetitions of the design may be performed by different judges or at different times. We shall discuss, in a subsequent section, the analysis when true ratings π_{1u},..., π_{tu} exist in the uth of y groups not necessarily identical from group to group

- no. of judge not necessary = no. of group, same kind of judges can form into 1 group

- ref：7. THE COMBINATION OF EXPERIMENTS ,where we discuss experiment is performed in groups of repetition of sizes n

Clarification

Group vs block vs repetition

- Block : is the fundamental unit of comparison where two treatments, i and j, are presented together.

  ref : Further definition follows with the assumption that, when treatment i appears with treatment j in a block, the probability that treatment i obtains top rating (or a rank of 1) is $\pi_i / (\pi_i + \pi_j)$.

  ref : One of the questions asked in connexion with this work pertains to the possibility of extending the analysis to incomplete block designs with larger block sizes. We are proceeding with a consideration of such extensions. The method of paired comparisons becomes inefficient where it is possible to rank more than two treatments at a time and where more than a few items or treatments are considered.

Group vs block vs repetition

- Repetition：A single complete running of the entire comparison design. This means enough comparisons are made to ensure all treatments have been compared and are connected within that data set.
- Ref ：and n will be reserved to denote the number of repetitions of the design when a repetition is defined to be a set of all pairs of treatments

Can we only have 1 repetition?

- Yes
- Ref : If three items are compared in a single repetition, the possible sets of rank sums are 2, 3, 4 and 3, 3, 3.
- But, the approximation appears to be fairly good for practical purposes if the number of repetitions is not too small (say $n > 15$).

How many judges can we use?

- No theoretical upper limit, but it is tied to the total number of repetitions n and number of groups g (n ≤ no. of judge, g ≤ no. of judge)
- n ≥ no. of judge, as we want to have all the data from a judge for every repetition
- g ≤ no. of judge, as if g were greater than the number of judges, the model would fail because some analytical groups would lack any corresponding data source.

Clarification

Example from paper

Table 4. *Rankings for two judges in the pork experiment*

Repetition ...	1			2			3			4			5		
	C	Cp	CP	C	Cp	CP	C	Cp	CP	C	Cp	CP	C	Cp	CP
Pair	Judge 1														
C, Cp	2	1	—	2	1	—	2	1	—	2	1	—	2	1	—
C, CP	2	—	1	1	—	2	2	—	1	2	—	1	2	—	1
Cp, CP	—	2	1	—	1	2	—	2	1	—	1	2	—	2	1
	Judge 2														
C, Cp	2	1	—	2	1	—	1	2	—	1	2	—	1	2	—
C, CP	1	—	2	1	—	2	1	—	2	1	—	2	2	—	1
Cp, CP	—	1	2	—	1	2	—	2	1	—	2	1	—	1	2

Example from paper

1. Block

Example: When a judge compares the Corn ration (C) pork against the Corn plus Peanut supplement (Cp) pork, that single comparison is a block.

2. Repetition (重複)

Example: Completing the comparison of C vs Cp, C vs CP, and Cp vs CP forms one repetition.

In the paper, each judge performed five repetitions (n=5).

3. Group (組別)

Example: Judge 1 is Group 1, and Judge 2 is Group 2. They are separate groups because their true ratings may be different.

2025-10-21

RANK ANALYSIS OF INCOMPLETE BLOCK DESIGN

THE METHOD OF PAIRED COMPARISON

Bradley, R. A., & Terry, M. E. (1952). Rank Analysis of Incomplete Block Designs: I. The Method of Paired Comparisons. *Biometrika*, *39*(3/4), 324–345. https://doi.org/10.2307/2334029

Why develop BT

- Overcame distributional and structural limitations
- Earlier Method：

1. assumptions of normality and equal variances; estimation is mathematically complex
2. depends on sensation
3. No explicit probabilistic model; cannot estimate underlying “true scores.”

The setting

1. "t" treatments (item, or suggestion in our case)
2. Treatments have true ratings $\pi_1,\ldots,\pi_t$
3. Every $\pi_i \geq 0$ and that $\sum \pi_i = 1$ (for convenience)
4. When comparing i to j, probability that treatment i obtain top rating (win) is $\pi_i / (\pi_i + \pi_j)$
5. r_{ijk} is rank of the ith treatment in the kth repetition of the block in which treatment i vs treatment j, $r_{ijk} = 3 - r_{jik}$(rank for win = 1, lose = 2)
6. p_1 ,..., p_t are estimates of $\pi_1,\ldots,\pi_t$
7. n is number of repetitions of the design when a repetition is defined to be a set of all pairs of treatments

*why the title said incomplete block design but they said "a set of all pairs of treatments" in definition of n?

Each block only observes a subset of treatments. However, when all blocks are aggregated, the overall data set may still represent one or more complete repetitions of all possible treatment pairs.The parameter n thus denotes how many full sets of pairwise comparisons are represented in total.

The setting 2

- true ratings π_{1u}, …, $\pi_{\square u}$ exist in the uth of y groups not necessarily identical from group to group

THE LIKELIHOOD FUNCTION - L

- Assuming probability independence between blocks or pairs of treatments
- When we are comparing i and j, the probability for i or j win will be

$$\left(\frac{\pi_i}{\pi_i+\pi_j}\right)^{2-r_{ijk}}\left(\frac{\pi_j}{\pi_i+\pi_j}\right)^{2-r_{jik}}$$

- Note: r_{ijk} is rank of the ith treatment in the kth repetition of the block in which treatment i vs treatment j, $r_{ijk} = 3 - r_{ijk}$ (rank for win = 1, lose = 2)
- So, when i win, it becomes $\pi_i / (\pi_i + \pi_j)$, if j win it becomes $\pi_j / (\pi_i + \pi_j)$
- Repeat for all treatment and multiply them together , we get the likelihood function

$$L = \prod_i \pi_i^{2n(t-1)-\sum_{j\neq i}\sum_k r_{ijk}} \prod_{i<j} (\pi_i+\pi_j)^{-n}.$$

LIKELIHOOD RATIO TESTS AND ESTIMATION

- Their null hypothesis(H_0): $\pi_i = 1/t$ $(i = 1,\ldots,t)$
- Alternative hypothesis (H_α): $\pi_i = \pi(h)$ $(h = 1,\ldots,m)$;
  $i = s_{h-1} + 1,\ldots,s_h$ where
  $s_0 = 0$, $s_m = t$ and
  $\sum(s_h - s_{h-1})\pi(h) = 1$
- where treatments have identical ratings within each of m groups of treatments while the groups themselves may differ
- But our case is a special case with
  H1: no π_i is assumed equal to any π_j $(i \neq j)$.
  Meaning that $m = t$ in H_α

LIKELIHOOD RATIO TESTS AND ESTIMATION 2

- If p(h) is the maximum-likelihood estimate of π(h), we have

$$\frac{a_i}{p_i} - n \sum_{j \neq i} (p_i + p_j)^{-1} = 0 \quad (i, j = 1, \ldots, t),$$

and

$$\sum_i p_i = 1.$$

We define

$$a_i = 2n(t-1) - \sum_{j \neq i} \sum_{k=1}^{n} r_{ijk}.$$

- The test statistic become

$$B_1 = n \sum_{i<j} \log (p_i + p_j) - \sum_i \left\{ 2n(t-1) - \sum_{j \neq i} \sum_{k=1}^{n} r_{ijk} \right\} \log p_i.$$

Table 4. *Rankings for two judges in the pork experiment*

Repetition ...	1			2			3			4			5		
	C	Cp	CP	C	Cp	CP	C	Cp	CP	C	Cp	CP	C	Cp	CP
Pair	Judge 1														
C, Cp	2	1	—	2	1	—	2	1	—	2	1	—	2	1	—
C, CP	2	—	1	1	—	2	2	—	1	2	—	1	2	—	1
Cp, CP	—	2	1	—	1	2	—	2	1	—	1	2	—	2	1
	Judge 2														
C, Cp	2	1	—	2	1	—	1	2	—	1	2	—	1	2	—
C, CP	1	—	2	1	—	2	1	—	2	1	—	2	2	—	1
Cp, CP	—	1	2	—	1	2	—	2	1	—	2	1	—	1	2

Things to decide

- Pooled analysis
- Combined analysis
- Affect how B_1 is calculated

To Be Done

- Pooled analysis / Combined analysis
- LARGE SAMPLE DISTRIBUTION
- VS my report last time

2025-10-13

ELO VS BT - Conceptual Part

	ELO	**BT**
purpose	Shows the real-time skill change using dynamic score	Estimate the static latent ability from all pairwise outcomes statistically
Nature of model	A system that learns step by step, updating scores after every match based on how players perform.	A probabilistic model based on maximum likelihood estimation (MLE).
Data requirement	Works incrementally, the sequence of the match results matters.	Requires the full set of pairwise comparisons or a **connected network**, no problem from sequence.

ELO VS BT - Conceptual Part

	ELO	**BT**
Assumption	Win probability depends on rating difference.	Win probability depends on the logit difference of latent ability.
Academic Difference	A heuristic updating rule (rule of thumb),designed for **practical use rather than for formal statistical inference.**	Grounded in probability theory and estimated through maximum likelihood estimation (MLE), making its parameters **interpretable and academically defensible**.

ELO VS BT Math Part

In the Bradley–Terry model, the derivative of the log-likelihood with respect to each player's ability β_i turns out to be $S_{ij} - P_{ij}$, where S_{ij} is the actual result (1 if player i wins, 0 otherwise) and P_{ij} is the predicted win probability. This means the model updates each player's score in the direction of the difference between actual and expected outcomes — if someone wins more than expected, their score increases, and if they lose more than expected, it decreases. This update rule is exactly what the Elo system uses, just applied one match at a time.

S_{ij} : i = player we are interested in; j = comparative player;
S = score of comparison (player i wins vs. player j: 1, 0 otherwise)

P_{ij} : probability (between 0.00 and 1.00) of player i winning over player j

What is a BIBD（Balanced Incomplete Block Design） for BWS

A scientific blueprint for survey design.
We often need to rank numerous items.
But, asking respondents to rank all items at once imposes a significant cognitive burden and results in poor data quality. The BWS solution is to present respondents with only a small subset of items at a time (a "block") and ask them to choose the best and worst from that subset.

What is a BIBD

BIBD’s role is to provide a systematic and unbiased way to construct these small subsets. Its ultimate goal is to ensure the "fairness" and "balance" of the overall survey design. This prevents any item from gaining an unfair advantage or disadvantage due to poor design, thereby ensuring the final preference ranking is reliable and valid.

What is a BIBD

A method rooted in the field of Experimental Design, used to systematically structure comparison tasks to collect the most accurate data with the highest efficiency.

Core Assumptions

1. Choice Reflects True Preference: What they choose is what they mean.The choice is not a random act.
2. Independence of Choices：Respondent's choice in each block is made independently.
3. No Context Effect: The appeal of an item is independent and does not fundamentally change based on the other "competing" items it is presented with.

Parameter	Term	Meaning	Explanation in the Context of a BWS Survey
v	varieties / points	Total number of items	The total number of things we want to rank (e.g., 10 items).
k	block size	Block size	The number of options presented to the respondent in each single question (e.g., 4 items are shown per task).
b	blocks	Total number of blocks	The total number of unique questions (or sets) in the entire survey design.
r	replication number	Replication number	The number of times each individual item appears across the entire survey. In a BIBD, this number is identical for all items.
λ (lambda)	Lambda	Pairwise co-occurrence	The number of times any given pair of distinct items appears together in the same question. In a BIBD, this number is also identical for all possible pairs.

Summary

1. A perfect BIBD provides the mathematically rigorous blueprint for our study.
2. It allows us to take our
3. v total items (varieties) and systematically organize them into b unique questions, or blocks.
4. Crucially, each of these blocks contains a smaller, manageable subset of k items for the respondent to evaluate.
5. The true power of this design is that by following this structure, we can guarantee two levels of perfect fairness: every single item appears exactly r times throughout the entire survey (replications), and every possible pair of items appears together in a question exactly λ times (Lambda).
6. It is this complete balance, defined by all five parameters, that gives us confidence that our final results will reflect true preferences, free from design-induced bias.

How does Elo get around connectedness problem that BT and BWS are worried about?

The problem still exist, If there is no connection at all, Elo cannot infer global rankings.
But as long as there is any minimal connection between groups, Elo will still work—
it will converge correctly, only more slowly as connectivity weakens.

Olesker-Taylor, Sam & Zanetti, Luca. (2024). An Analysis of Elo Rating Systems via Markov Chains. 10.48550/arXiv.2406.05869.

Others major pairwise comparison methods

1. Analytic Hierarchy Process (AHP)：likert scale + weighted result

2. PROMETHEE（Preference Ranking Organization METHod for Enrichment Evaluations）：Compares alternatives pairwise under each criterion, converts performance differences into preference intensities, and aggregates them into an overall outranking flow to rank the options.

2025-10-07

ELO: score & rank

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}}.$$

$$E_B = \frac{1}{1 + 10^{(R_A - R_B)/400}}.$$

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$$R'_A = R_A + K(S_A - E_A).$$

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E_A prob of a win

R_A current ELO score of player A

R'_A ELO score after the match

K-factor one of the factor that determine how violently the score change, usually high if the player have high **uncertainty** (new players, old player who didn't play for a while)

- We can use K factor= 40 to make it converge quickier
- S_A: game score (lose=0, tie=0.5, win=1)

Starting Match:

- Everyone starts with baseline 1500 in chess, but actually it doesn't matters

What is the min “N” comparison needed?

- There is no certain “N”, but we can do it until the score become fixed.

About video on ELO for smartphone comparisons

No math details, but similar to our situation, the testing subject (picture showed) didn't change at all.
They give all the participant exact same picture to compare with and the “experiment” only consist one round.
It means they didn't “give every testing subject a second chance”, similar to us as the picture used in the test is same and never change, thus no uncertainty from the picture showed to participant.

Why 400？

Elo suggested a scaling constant such that **200 points difference in favor of player A corresponds to a 0.75 expected score for A**.

- 200 points corresponds to chess levels (masters?)

400 is an empirical scaling factor in the Elo formula, ensures the formula matches observed win probabilities (e.g., 200-point difference –

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}}.$$

$$E_B = \frac{1}{1 + 10^{(R_A - R_B)/400}}.$$

CSDN @软件的小哥哥

Pros and cons

Pros:

- Simple and Effective
- Ability to predict the expected outcome of a match based on the rating difference between two players

Cons:

- Cannot accurately assess individual contributions in *team-based games*
- *High-ranked players are incentivized not to play against lower-ranked opponents* to avoid the risk of losing a large number of rating points from a single upset

Example

- Ranks scientific journals by adapting the Elo rating system
- Lehmann, R., & Wohlrabe, K. (2017). Who is the 'Journal Grand Master'? A new ranking based on the Elo rating system. *Journal of Informetrics*, *11*, 800–809. https://doi.org/10.1016/j.joi.2017.05.004

Pros and cons - BT

Pros

1. **Interpretability.**
   The model is $P(i > j) = \text{logit}^{-1}(\beta_i - \beta_j)$

   So β is a latent “ability.”

   A one-unit gap $(\beta_i - \beta_j = 1)$

   means i beats j with ≈73% probability—making rankings and upset probabilities easy to explain.

2. **Works with incomplete/uneven comparisons.**
   Don’t have to compare every pair of items; MLE uses whatever pairwise data we have as long as the comparison graph is (strongly) connected, so it’s practical for real-world

Pros and cons - BT

cons

1. **Strong assumptions**
   The BT model assumes each outcome is independent and context-free, but real data often show context effects

2. **Estimation pathologies with separation/sparsity**
   If an item never loses (or never wins) or the comparison graph isn't well connected, the MLE may not exist or be unstable.

   Typical fixes include adding regularization/Bayesian priors, constraints, or explicit tie/home-field extensions.（thx gpt）

BWS - how if the item is never the best or worst ?

As long as the BWS design is **well connected**, every item can be indirectly compared through others, so it's not a problem if some items are never chosen as best or worst.

A well-designed **Balanced Incomplete Block Design** (BIBD) ensures that:

- each item appears several times across different sets;
- each item co-occurs with other items roughly the same number of times;

therefore, even if an item is never directly chosen as the best or worst, it can still be positioned on the overall preference scale through indirect (transitive) comparisons.

Pros and cons - BWS

Pros

Produces ratio-scaled, discrimination-friendly data, allowing clear comparisons between items.

Reduces response bias found in rating scales by forcing trade-offs (best vs. worst choices).

Cons

Cognitively demanding when many items are compared or sets are large.

Measures are relative, not absolute — results depend on the specific items included.

Example - BWS

- Applied it to a small set of items by measuring consumer preferences for six different weekend getaway destinations.
- ranking the perceived quality of 72 academic marketing journals.

- Louviere, J., Lings, I., Islam, T., Gudergan, S., & Flynn, T. (2013). An introduction to the application of (case 1) best-worst scaling in marketing research. *International Journal of Research in Marketing*, *30*(3), 292–303. https://doi.org/10.1016/j.ijresmar.2012.10.002

Rank Centrality

Stuck at stationary distribution…

2025-09-22

Scoring two content creation solutions

1. LLM (e.g., ChatGPT) do pairwise comparisons and provides scores
2. Human raters pairwise comparisons and provides scores
3. Human raters do pre-rating and given qualitative feedback on why they prefer some solutions over others — we can then feed these qualitative assessments into an LLM to be an automated rater
   a. Un-prompted LLM might have different preferences than human raters (e.g., structure > content)

ELO: score & rank

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}}.$$

$$E_B = \frac{1}{1 + 10^{(R_A - R_B)/400}}.$$

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$$R'_A = R_A + K(S_A - E_A).$$

CSDN @学软件的小垃圾

E_A prob of a win

R_A current ELO score of player A

R'_A ELO score after the match

K-factor one of the factor that determine how violently the score change, usually high if the player have high **uncertainty** (new players, old player who didn't play for a while)

- We can use K factor= 40 to make it converge quickier
- S_A: game score (lose=0, tie=0.5, win=1)

Starting Match:

- Everyone starts with baseline 1500 in chess, but actually it doesn't matters

What is the min “N” comparison needed?

- There is no certain “N”, but we can do it until the score become fixed.

Bradley-Terry model

- Calculate hidden “power value” p_i, using the existing comparison result and Maximum Likelihood estimation method

- So, how we find out each person’s p?
  Step
  1）Recording all the comparison result
  2）Use MLE to estimate each player’s ability parameter p_i based on the observed match outcomes, such that the likelihood of these outcomes under the model is maximized.
  (We can use Python choix or R BradleyTerry2)

Bradley-Terry model

The Math part

- Prob. i win j is calculated by $P(i>j) = p_i/(p_i+p_j)$
- Let $\beta_i = \ln(p_i)$. Thus $p_i = e^{\beta_i}$
- So, $P(i > j) = e^{\beta_i} / (e^{\beta_i} + e^{\beta_{\square}})$
- We define odds as $Odds(i > j) = P(i > j) / P(j > i)$
- Thus $Odds(i > j) = [p_i / (p_i + p_{\square})] / [p_{\square} / (p_i + p_{\square})]$
- $Odds(i > j) = p_i / p_{\square}$
- $\ln(Odds(i > j)) = \ln(p_i / p_{\square})$
- $\ln(Odds(i > j)) = \ln(p_i) - \ln(p_{\square})$

Now！ the hard part

- We know for a single comparison, $P(i>j) = p_i/(p_i+p_j)$
- If i beat j for w_{ij} times, and j win i for w_{ji} times, the prob. that we will get the result mentioned will be:
  $L = [p_i / (p_i + p_{\square})]^{w_{i\square}} * [p_{\square} / (p_i + p_{\square})]^{w_{\square i}}$
  Where L is Likelihood Function for observing i beat j for w_{ij} times, and j win i for w_{ji} times

Bradley-Terry model

The Math part

- Expanding this mindset to other's player result, we can say:
  $L(p_1, p_2, \ldots \mid data) = \Pi\ (p_i / (p_i + p_{\square}))^{w_{i\square}}$
- Using ln()
  $\ln(L) = \ln [\ \Pi\ (p_i / (p_i + p_{\square}))^{w_{i\square}}\ ]$
  $\ln(L) = \Sigma\ [\ w_{i\square} * \ln(p_i / (p_i + p_{\square}))\ ]$
  $\ln(L) = \Sigma\ [\ w_{i\square} * (\ln(p_i) - \ln(p_i + p_{\square}))\ ]$
- Finally, the not for human part
  We need to max. ln(L)
  By finding all the partial derivatives ☺

So…. thx python and R ☺

Bradley-Terry model

What is the min “N” comparison needed?

Still cant find reliable resource for this

~~AI: 15~20, but inconsistent because it also said 10~20~~

Best Worst Scaling (BWS)

- Raters compare between **3 or more options** and choose the **best and the worst** from it and do several groups of comparisons
- Calculate how many times each option was voted as best and as worst
- Subtract the number of worst votes from the number of best votes for each option, and then rank the options by that result

- **Thurstone – only compare 2 options, rank by win rate

BWS

- What is the min “N” comparison needed?
  No absolute number that is needed for this
  But need to follow:
  a)**Item Frequency Balance**: Each item is presented to respondents the same number of times, or as close as possible, across the entire survey design (summed across all versions).
  b)**Pairwise Balance**: Any two items (e.g., "Price" and "Quality") appear together in the same set the same number of times, or as close as possible, as any other pair of items (e.g., "Price" and "Ambiance").

BWS

- What is the min “N” comparison needed?
  c)Ensure the design has "**Connectivity**." This means that even if item A and item B never appear in the same set together, the model can still indirectly infer the relative importance between A and B through their respective comparisons with other items like C, D, and E.

The “I still dont really can explain part”

Rank Centrality

- Assigns global scores to items from pair-wise comparison data
- Using **random walk** on a comparison data, where an item‘s final score is its stationary probability
- In short :using the win rate of the option to distribute a fix total score multiple times until the result converged

  1）Build a matrix using the outcome of comparison

  2）Create a probability matrix that describes the flow of "rank" from losers to winners.

  3）Repeatedly update a score vector (s_new = M * s_old) until it stabilizes. This process finds the principal eigenvector of the matrix.

  4）The final, stable score vector provides the ranks. The higher the score, the higher the rank.

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Scoring two content creation solutions

1. LLM (e.g., ChatGPT) do pairwise comparisons and provides scores
2. Human raters pairwise comparisons and provides scores
3. Human raters do **pre-rating** and given **qualitative feedback** on why they prefer some solutions over others — we can then feed these qualitative assessments into an LLM to be an automated rater
   a. Un-prompted LLM might have different preferences than human raters (e.g., structure > content)

pairwise comparison

determine their relative importance

ELO: score & rank

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}}.$$

$$E_B = \frac{1}{1 + 10^{(R_A - R_B)/400}}.$$

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$$R'_A = R_A + K(S_A - E_A).$$

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E_A prob of a win

R_A current ELO score of player A

R'_A ELO score after the match

K-factor one of the factor that determine how violently the score change, usually high if the player have high **uncertainty** (new players, old player who didn't play for a while)

- S_A: score (0, 0.5, 1)

Starting Match:

- Everyone starts with baseline 1500 in chess

Bradley-Terry model

- Calculate hidden “power value” p_i, using the existing comparison result and Maximum Likelihood estimation method

$$P(i > j) = p_i / (p_i + p_j)$$

$$\ln(P(i > j) / P(j > i)) = \beta_i - \beta_j$$

$$\beta_i = \ln(p_i)$$

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BWS

- Participant compare between **3 or more options** and chose the best and the worse from it and do severe groups of comparison
- Calculate how many times each option was voted as best and as worst
- Subtract the number of worst votes from the number of best votes for each option, and then rank the options by that result

- **Thurstone – only compare 2 options, rank by win rate